

RB-006

RB-006: Wind Interaction and Cross-Flow Effects

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Outdoor Ventilation Standard

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Revision history

v1.1 — 2026-09-26. Errata:

- Section 3.2 / Tables 3.2a-h: the u_o column regenerated from the Heskestad formula, following RB-001 Table 3.5 v1.1 (Gas Medium 2.30/2.12/1.99/1.88/1.71 → 2.21/2.09/1.99/1.91/1.78 m/s; other sources likewise), and all five deflection columns (2, 5, 8, 10 and 15 mph) regenerated from the linear formula $\Delta x = 0.35 * U_w * z / u_o(z)$ on that u_o (Gas Medium 10 mph: 0.33/0.47/0.63/0.81/1.24 → 0.33/0.46/0.60/0.74/1.07 m; 15 mph: 0.60/0.87/1.17/1.52/2.39 → 0.49/0.68/0.89/1.12/1.61 m; the 2-8 mph cells move by at most 0.15 m, Pellet Smoker Low 48" 8 mph 1.66 → 1.51 m; Charcoal Kettle 15 mph at 48" 2.83 → 1.61 m). The v1.0 10 and 15 mph cells carried an unstated partial application of the Section 3.1 Froude correction. Cells with $Fr > 2.7$ are marked [†] as the disrupted regime of Table 3.8, with a footnote under each table; the Section 3.2 introduction states that the linear formula is used throughout. The key observations under the tables (charcoal kettle at 30"/5 mph 19 → 20 inches; gas high-output at 30"/10 mph 20 → 19 inches; 15 mph floor 18 → 15 inches and 48" range 72-187 → 51-112 inches), Table 3.7 (gust design deflection: mean 25 → 23" and peak 42 → 40" at 10 mph; Fr 1.13/1.58/2.68/2.25/3.82 → 1.12/1.57/2.67/2.24/3.81) and the Section 3.9.5 overhang-per-mph table (1.8-4.2 instead of 2.3-4.0 inches per mph; 5 mph 9-21 instead of 11-20 inches) follow, as do the Froude quotes in Section 3.3 (gas high-output at 30"/15 mph Fr 2.75 → 2.72) and the Section 6 figure note (Fr 1.13 → 1.12 at 5 mph).
- Section 3.1: the Froude correction $(1 + 0.3 (Fr - 1))$ is stated as an upper-bound estimate for $Fr > 1$ that is not applied in Tables 3.2a-h.
- Section 3.3 / Table 3.3: u_o and $Fr = U_w / u_o$ regenerated on the v1.1 u_o for all four source blocks (Gas Medium 30": Fr 0.45/1.13/1.80/2.25/3.37 → 0.45/1.12/1.79/2.24/3.36; Charcoal Kettle 30": u_o 1.25 → 1.17 m/s, Fr 0.71/1.79/2.86/3.58/5.37 → 0.76/1.91/3.06/3.82/5.73; Gas High-Output and Pellet Smoker Low likewise); the Gas Medium block is identical to the site's rb-006 dataset CSV.
- Section 3.9.4 orientation tables (Gas Medium 30"): centerline-exit winds recomputed from $U_{crit} = OH * u_o / (0.35 z)$ with the v1.1 u_o — 57" x 53" hood 8.2/8.5/6.3 → 6.9/7.0/5.1 mph (orientation effect 4% → 2%, diagonal -23% → -26%); 66" x 55" hood 6.0/9.3 → 5.1/7.7 mph (+55% → +50%);

the key finding's "30 to 55%" → "30 to 50%".

- Section 3.4 / Tables 3.4a and 3.4b: the three critical-wind-speed columns re-derived from $U_{crit} = \Delta * u_o(z) / (0.35 * z)$ with the printed base-margin ($K = 1.38$) overhang and the regenerated u_o (Gas Medium 30": 6.7/9.7/12.6 → 4.8/7.0/9.3 mph; Gas High-Output 18": 8.0/12.5/16.7 → 6.9/10.9/15.0 mph; Charcoal Kettle 30": 5.0/6.7/8.4 → 3.7/5.1/6.6 mph; Pellet Smoker Low 30": 4.2/5.8/7.3 → 2.6/3.9/5.2 mph; all 40 rows likewise); v1.0 computed these columns with the wider $K = 1.70$ hood while labelling the base-margin overhang. The b_T cells for Gas Small, Wood-Fired and Pellet Smoker High are corrected from Gas Medium's values to their own (0.100 → 0.091, 0.098 and 0.091 m at 18", and so on). A basis note is added above Table 3.4a. The findings under the tables, the Section 3.11 answer to Key Question 1 (25% escape 6.7 → 4.8 mph; centerline exit 9.7 → 7.0 mph; practical thresholds 5/7/10 → 3/5/7 mph mean; weaker-source offsets 30-40% → 25-45% lower with thresholds 3/5/7 → 1.5-2/2-3/3-4 mph mean; high-output offsets 20-30% → 5-20% higher), Section 4.4 (11.1/9.7/8.1 → 9.8/7.0/5.4 mph; 37% → 82%; 0.5-0.8 → 0.4-0.9 mph per 6") and Section 4.5 (5.0/6.7 → 3.7/5.1 mph; 25-35% → 15-40% lower) follow.
- Section 4.3 / Table 4.3: column added mapping each wind-speed band to the RB-008 exposure class and the class-representative F_{wind} it adopts (Sheltered 1.3, Moderate 1.6, Exposed 2.5 or 1.8 with side panels); note added that the speed-resolved values apply when wind speed at cooking height is measured. The RB-008 paragraph in Section 4.7 is reworded accordingly. No numbers change.
- Section 3.1: drafting artifacts in the worked deflection example removed (editorial; no numbers change).

Every regenerated value is the output of the paper's printed formula with its printed inputs; the cell-by-cell ledger (old → new, with the computation behind each) is kept with the site source.

1. Topic Definition

This paper models the interaction between ambient wind and the **Buoyant Cooking Plume** generated by outdoor barbecue sources. Wind is the single most consequential environmental variable for outdoor hood capture performance. While RB-001 and RB-003 characterized the plume and derived CFM requirements in still-air and light-breeze conditions, the present paper addresses the full spectrum of **Wind-Affected Plume Behavior** from calm conditions through severe exposure at 15 mph.

The scope encompasses:

1. **Plume deflection modeling.** For each source type catalogued in RB-001 Table 3.1, calculation of lateral plume displacement at hood height as a function of wind speed (0, 2, 5, 8, 10, 15 mph) at five standard mounting heights (18", 24", 30", 36", 48"). The Briggs bent-plume trajectory equations

provide the primary analytical framework, supplemented by Froude number and momentum ratio characterization.

2. **Critical wind speed analysis.** Determination of the wind speed at which plume centerline exits a standard-width hood, the wind speed at which more than 25% of plume mass exits the **Capture Envelope**, and the wind speed at which capture becomes unreliable with more than 50% of the plume in the **Missed Plume Region**.
3. **Wind-enhanced entrainment.** Quantification of how crosswind increases the entrainment rate, widens the plume asymmetrically, and accelerates dilution via **Open-Boundary Dilution** at the hood height.
4. **Plume bifurcation and disruption.** Identification of wind speeds at which the plume develops counter-rotating vortex pair (CVP) structure, loses coherent column form, or becomes ground-hugging or fully disrupted.
5. **Mitigation strategy quantification.** Side panels, rear panels, increased CFM, hood orientation, and deeper overhang evaluated with quantitative effectiveness metrics and CFM-equivalent improvement values.
6. **Wind exposure classification.** A four-tier classification system (Sheltered, Moderate, Exposed, Severe) with specific design recommendations for each class.
7. **Gustiness and turbulence effects.** Distinction between steady-wind deflection, gust-driven peak deflection, and turbulence-induced capture reliability degradation.

Problem Framing

Outdoor cooking installations operate in a fundamentally different aerodynamic environment than indoor kitchens. Indoor hoods benefit from enclosure walls that block lateral airflow, ceiling surfaces that redirect escaped plume gas, and room pressurization that assists recirculation toward the exhaust. Outdoors, the **Buoyant Cooking Plume** rises into an atmosphere that is rarely still. Even on nominally calm days, thermal convection and terrain effects produce air movement of 1 to 3 mph at cooking-surface height. In moderately exposed locations (open patios, rooftop terraces, beachfront installations), sustained winds of 5 to 10 mph are routine. In exposed locations (hilltops, open fields, coastal sites), 10 to 15 mph winds occur frequently.

Wind interacts with the buoyant plume through three primary mechanisms:

Mechanism 1 — Lateral deflection. The horizontal wind momentum bends the rising plume trajectory downwind. The plume centerline follows a curved path rather than a vertical line. At hood height, the plume centerline is displaced laterally from its still-air position by a distance that increases

with wind speed and mounting height. If this displacement exceeds the hood's overhang on the downwind side, the plume core escapes.

Mechanism 2 — Enhanced entrainment and asymmetric spreading. Wind increases the velocity shear at the windward plume boundary, accelerating turbulent engulfment and increasing the entrainment rate above its still-air value. The plume widens asymmetrically: wider on the windward side (where shear-driven entrainment is enhanced) and narrower on the leeward side (where the ambient flow partially suppresses entrainment). The net effect is increased total entrainment, faster dilution, and a plume cross-section that is elliptical rather than circular.

Mechanism 3 — Structural disruption. At sufficient wind speed, the plume loses its coherent columnar structure. The rising buoyant gas column develops a counter-rotating vortex pair (CVP) — two vortex tubes aligned with the wind direction that wrap around the bent plume. At higher wind speeds, the CVP structure itself becomes unstable, and the plume breaks up into discrete puffs or is flattened against the cooking surface. In this regime, the concept of a capturable plume column no longer applies, and the hood functions as a wind shield rather than a plume interceptor.

The key dimensionless parameter governing which mechanism dominates is the **Froude number for crosswind**:

$$Fr = U_{wind} / W_{plume}$$

where U_{wind} is the ambient wind speed at hood height and W_{plume} is the plume centerline vertical velocity at hood height. When Fr is much less than 1, buoyancy dominates and the plume rises nearly vertically (Mechanism 1 produces small deflection). When Fr is of order 1, wind and buoyancy are competitive and the plume is significantly bent (all three mechanisms are active). When Fr is much greater than 1, wind dominates and the plume is laid over or disrupted (Mechanism 3 prevails).

This paper quantifies the plume behavior across the full Froude number range relevant to outdoor cooking: from Fr approximately 0 (still air) through Fr approximately 3 to 5 (severe wind disruption).

Relationship to Foundation Papers

RB-001 provides the source parameters (Q_c , z_o , D_{eff}) and still-air plume properties (u_o , ΔT_o , b_T) at each height. RB-003 provides the still-air CFM requirements, momentum flux values, and edge capture velocities. This paper extends both into the wind-exposed domain by computing the modified plume trajectory, enhanced entrainment, and additional CFM and geometric requirements imposed by wind.

The preliminary wind correction factors used in RB-003 ($F_{\text{wind}} = 1.3$ for standard outdoor, $F_{\text{wind}} = 1.6$ for sustained light wind) are refined and replaced by wind-speed-specific correction factors derived from the detailed analysis in this paper.

2. Relevant Physical Principles

2.1 Briggs Plume Rise and Bent-Plume Trajectory Equations

Gary Briggs developed the most widely applied framework for predicting the trajectory of a buoyant plume in a crosswind, originally for atmospheric dispersion modeling of power plant and industrial stack plumes (Briggs, 1969, 1975, 1984). The Briggs formulation treats the plume as a continuous bent column that rises vertically due to buoyancy while being advected horizontally by the ambient wind.

Coordinate system. The plume source is at the origin. The x-axis points downwind (in the direction of the ambient wind). The z-axis points vertically upward. The plume centerline trajectory is described by the function $z(x)$ — the height of the plume centerline as a function of downwind distance.

Briggs two-thirds law. For a buoyant plume in a uniform crosswind with negligible ambient stratification (appropriate for the atmospheric boundary layer at cooking heights), the plume rise is:

$$\Delta_h(x) = 1.6 * F_b^{(1/3)} * x^{(2/3)} / U_w$$

where $\Delta_h(x)$ is the plume rise above the source at downwind distance x , F_b is the buoyancy flux parameter (m^4/s^3), and U_w is the mean wind speed (m/s).

The buoyancy flux parameter F_b is defined as:

$$F_b = g * Q_c / (\pi * \rho_{\text{inf}} * c_p * T_{\text{inf}})$$

Substituting standard ambient conditions ($g = 9.81 \text{ m/s}^2$, $\rho_{\text{inf}} = 1.20 \text{ kg/m}^3$, $c_p = 1000 \text{ J/(kg}\cdot\text{K)}$, $T_{\text{inf}} = 293 \text{ K}$):

$$F_b = 9.81 * Q_c / (\pi * 1.20 * 1000 * 293) = Q_c / 112,700 = 8.87 \times 10^{(-6)} * Q_c \text{ [m}^4/\text{s}^3, \text{ with } Q_c \text{ in watts]}$$

For Q_c in kilowatts:

$$F_b = 8.87 \times 10^{-3} \times Q_c [m^4/s^3]$$

Plume trajectory inversion. For hood capture analysis, the relevant question is not "how high does the plume rise at downwind distance x ?" but rather "how far downwind has the plume drifted by the time it reaches hood height z ?" Inverting the Briggs equation:

$$x(z) = (z \times U_w / (1.6 \times F_b^{(1/3)}))^{(3/2)}$$

This gives the downwind displacement of the plume centerline at height z above the source. This is the lateral deflection that the hood must accommodate.

However, for outdoor cooking plumes, there is a subtlety: the plume source is not a point at $z = 0$; it is a distributed area source with a virtual origin at z_0 (which is negative, below the cooking surface). The plume already has significant upward momentum at the cooking surface. The Briggs formulation assumes a point source releasing buoyancy flux F_b into a crossflow, with the plume initially horizontal. For a cooking plume that is already rising vertically, the deflection is less than the Briggs formula predicts for low heights and converges toward the Briggs prediction at greater heights.

To account for this, the deflection is computed using an effective trajectory that begins at the virtual origin with the still-air plume velocity:

$$\Delta x(z) = \text{integral from } 0 \text{ to } z \text{ of } [U_w / u_0(z')] dz'$$

where $u_0(z') = 1.03 \times Q_c^{(1/3)} \times (z' - z_0)^{(-1/3)}$ is the still-air centerline velocity at height z' . This integral expresses the physical principle that at each height increment dz' , the plume takes time $dt = dz'/u_0$ to rise through that increment, and during that time the wind advects the plume horizontally by $dx = U_w \times dt$. The total horizontal displacement is the cumulative sum of these incremental advections.

Evaluating the integral:

$$\Delta x(z) = U_w \times \text{integral from } 0 \text{ to } z \text{ of } [(z' - z_0)^{(1/3)} / (1.03 \times Q_c^{(1/3)})] dz'$$

$$\Delta x(z) = \left(U_w / (1.03 * Q_c^{1/3}) \right) * (3/4) * [(z - z_0)^{4/3} - (-z_0)^{4/3}]$$

This is the **primary deflection equation** used throughout this paper. It gives the lateral displacement of the plume centerline at height z above the cooking surface due to a uniform crosswind of speed U_w .

2.2 Froude Number and Wind Momentum Ratio

Two dimensionless parameters characterize the relative strength of wind forcing versus plume buoyancy:

Crosswind Froude number:

$$Fr(z) = U_w / u_0(z) = U_w / (1.03 * Q_c^{1/3} * (z - z_0)^{-1/3})$$

$$Fr(z) = U_w * (z - z_0)^{1/3} / (1.03 * Q_c^{1/3})$$

This is a height-dependent quantity that increases with height (because u_0 decreases) and with wind speed. The Froude number governs the plume trajectory shape:

Fr Range	Regime	Plume Behavior
$Fr < 0.3$	Buoyancy-dominated	Near-vertical rise; small deflection; plume retains circular cross-section
$0.3 < Fr < 1.0$	Transitional	Significant bending; elliptical cross-section; counter-rotating vortex pair developing
$1.0 < Fr < 3.0$	Wind-dominated	Strongly bent plume; CVP fully developed; plume may not reach intended hood height vertically above source
$Fr > 3.0$	Disrupted	Plume laid over or broken into puffs; coherent column structure lost

Wind momentum ratio (velocity ratio squared):

$$R = (\rho_p * W_0^2) / (\rho_{inf} * U_w^2)$$

where ρ_p is the plume density, W_o is the plume vertical velocity, ρ_{inf} is the ambient density, and U_w is the wind speed. In the Boussinesq approximation (ρ_p approximately equals ρ_{inf}):

$$R \text{ approximately equals } (W_o / U_w)^2 = 1 / Fr^2$$

When $R \gg 1$, the plume momentum dominates and the plume rises through the crossflow with minimal deflection. When $R \ll 1$, wind momentum dominates and the plume is swept downwind. The critical threshold $R = 1$ (equivalently $Fr = 1$) marks the transition between buoyancy-dominated and wind-dominated regimes.

2.3 Wind-Enhanced Entrainment

In still air, the entrainment coefficient for the far-field buoyant plume is $\alpha_o = 0.11$ (Gaussian profile, SFPE Handbook convention, as established in RB-001 and RB-002). In a crosswind, the effective entrainment coefficient increases due to two mechanisms:

Shear-enhanced engulfment. The wind creates additional velocity shear at the windward plume boundary, increasing the rate of Kelvin-Helmholtz instability development and large-eddy engulfment. The enhancement scales with the ratio of wind speed to plume velocity:

$$\alpha_{wind} = \alpha_o * (1 + \beta * Fr)$$

where β is an empirical coefficient. Experimental data from Hewett et al. (1971), Fackrell & Robins (1982), and Davidson (1986) suggest $\beta = 0.5$ to 1.0 for round plumes in crossflow.

Turbulent diffusion from ambient. The ambient wind carries turbulent eddies at scales ranging from the atmospheric boundary layer thickness (hundreds of meters) down to small-scale eddies comparable to the plume diameter. Eddies at the plume scale enhance mixing at the plume boundary. The turbulence intensity $I_u = u_{rms} / U_w$ is typically 0.15 to 0.30 in the atmospheric surface layer at heights of 1 to 2 meters. This adds an additional entrainment contribution:

$$\alpha_{turb} = \alpha_o * C_t * I_u * Fr$$

where C_t is of order 0.5 to 1.0 .

The combined effective entrainment coefficient in crosswind is:

$$\alpha_{eff} = \alpha_0 * (1 + (\beta + C_t * I_u) * Fr)$$

For the purpose of this analysis, the following simplified model is adopted:

$$\alpha_{eff} = \alpha_0 * (1 + 0.7 * Fr)$$

This produces:

Fr	α_{eff}	Ratio to still-air
0 (still air)	0.11	1.00
0.3 (light breeze)	0.13	1.21
0.5 (moderate breeze)	0.15	1.35
1.0 (strong breeze)	0.19	1.70
2.0 (high wind)	0.25	2.30
3.0 (severe wind)	0.34	3.10

At $Fr = 1.0$, the entrainment rate is 70% higher than still air. At $Fr = 3.0$, it has more than tripled. This enhanced entrainment causes the plume to expand faster, dilute more rapidly, and lose thermal coherence at lower heights than in still air.

2.4 Counter-Rotating Vortex Pair (CVP) Structure

When a buoyant plume or jet issues vertically into a crossflow, the interaction between the vertical momentum of the plume and the horizontal momentum of the wind generates a pair of counter-rotating vortices within the bent plume cross-section. This counter-rotating vortex pair (CVP) is a fundamental feature of jets and plumes in crossflow, extensively documented in the fluid dynamics literature (Fric & Roshko, 1994; Mahesh, 2013; Karagozian, 2014).

The CVP develops through the following mechanism:

1. The wind flows around the rising plume column, creating a pressure differential between the windward (high pressure) and leeward (low pressure) sides.
2. This pressure differential drives a secondary flow within the plume: ambient air wraps around the plume from windward to leeward, creating two vortex tubes aligned with the wind direction.

3. The vortices entrain plume fluid from the core and transport it to the plume periphery, while simultaneously drawing ambient air into the plume interior.
4. The result is a kidney-shaped or bifurcated plume cross-section, with two concentration maxima (one in each vortex) rather than a single Gaussian peak.

Onset of CVP structure. The CVP begins to develop at Fr approximately 0.5 and is fully established by Fr approximately 1.0. Below Fr approximately 0.3, the plume cross-section remains approximately circular (Gaussian).

Implications for hood capture. The CVP structure has two consequences for capture:

1. **Wider effective plume.** The CVP spreads the contaminant-laden plume fluid over a wider area than the equivalent Gaussian plume, increasing the required capture width.
2. **Concentration bifurcation.** The plume splits into two lobes. If one lobe extends beyond the hood boundary while the other remains within it, the hood captures only half the contaminant, even though it may appear to be intercepting the plume center.

For practical outdoor cooking analysis, the CVP becomes relevant at wind speeds where Fr exceeds approximately 0.5. Using the plume velocities from RB-001 Table 3.5 (1.0 to 2.8 m/s at 18 inches; 1.0 to 2.1 m/s at 48 inches), the CVP onset wind speed is:

$$U_{w_CVP} = 0.5 * u_0(z)$$

At 30 inches: $U_{w_CVP} = 0.5 * 1.0$ to $0.5 * 2.4 = 0.5$ to 1.2 m/s (1.1 to 2.7 mph)

This means that even light winds can initiate CVP development in weaker plumes. The cooking plume in crosswind is rarely a simple bent Gaussian column; it more commonly exhibits some degree of vortex-pair structure.

2.5 Plume Disruption and Incoherence Thresholds

At sufficiently high Froude numbers, the plume loses coherent columnar structure entirely. Three disruption regimes are identified:

Regime 1 — Coherent bent plume ($Fr < 1.5$). The plume maintains a continuous, identifiable column that can be tracked from source to hood height. The Briggs trajectory equations apply. The plume is capturable by a properly positioned hood.

Regime 2 — Intermittent coherence ($1.5 < Fr < 3.0$). The plume alternates between coherent column structure and disrupted, puff-like behavior. Wind gusts periodically break the column into discrete thermals that are advected downwind. During coherent periods, the plume reaches the hood; during disrupted periods, it does not. Time-averaged capture efficiency falls below 50%. The Briggs trajectory provides an estimate of the mean plume path but does not capture the intermittent failures.

Regime 3 — Full disruption ($Fr > 3.0$). The plume cannot maintain columnar structure. The buoyant gas is immediately swept downwind upon release, forming a thin, spreading layer rather than a rising column. The plume may become ground-hugging as the wind prevents vertical rise. In this regime, the concept of a rising plume intercepted by an overhead hood no longer applies. Capture, if achievable, requires enclosure of the cooking area rather than overhead extraction.

For outdoor cooking plumes, Regime 3 ($Fr > 3.0$) corresponds to wind speeds of:

$$U_{w_disruption} = 3.0 * u_0(z)$$

At 30 inches: $U_{w_disruption} = 3.0 * 1.0$ to $3.0 * 2.4 = 3.0$ to 7.3 m/s (6.7 to 16.3 mph)

This means that for the weakest plumes (pellet smoker low, charcoal kettle), full disruption can occur at wind speeds as low as 7 mph, while the strongest plumes (gas grill high-output) can remain coherent up to approximately 16 mph at 30 inches.

2.6 Gust Factor and Turbulence Intensity

Ambient wind is not steady. It fluctuates on timescales of seconds to minutes. The relationship between the mean wind speed and the peak instantaneous wind speed is characterized by the gust factor:

$$G = U_{peak} / U_{mean}$$

In the atmospheric surface layer at heights of 1 to 3 meters above ground (the relevant height range for outdoor cooking installations), measured gust factors typically range from 1.5 to 2.0, depending on terrain roughness and atmospheric stability:

Terrain	Typical Gust Factor G
Urban / suburban (buildings, trees, fences)	1.5 - 1.7
Open suburban (scattered obstacles)	1.6 - 1.8
Open terrain (fields, flat ground)	1.7 - 2.0
Coastal / hilltop (fully exposed)	1.8 - 2.0

For this analysis, a gust factor of $G = 1.7$ is used as the standard value, with $G = 2.0$ for exposed locations.

The practical implication is that a site with a mean wind speed of 5 mph experiences instantaneous gusts of 8.5 mph ($G = 1.7$) or 10 mph ($G = 2.0$). The plume responds to instantaneous wind speed on a timescale comparable to the plume rise time (approximately z / u_0 , typically 0.3 to 1.0 seconds for cooking plumes at standard heights). Since gusts at 1 to 3 Hz are within the plume response bandwidth, the plume deflection during a gust corresponds to the instantaneous wind speed, not the mean.

This means that the critical wind speed for capture failure should be compared against the gust speed, not the mean speed. A site with a mean wind of 5 mph and $G = 1.7$ experiences gust-equivalent deflections corresponding to 8.5 mph. The capture system must be designed for the gust deflection, not the mean deflection.

Turbulence intensity $I_u = \sigma_u / U_{\text{mean}}$ (the ratio of the standard deviation of wind speed fluctuations to the mean wind speed) is typically 0.15 to 0.30 in the surface layer:

Terrain	Turbulence Intensity I_u
Urban / suburban	0.25 - 0.35
Open suburban	0.20 - 0.25
Open terrain	0.15 - 0.20
Coastal / hilltop	0.10 - 0.15

Higher turbulence intensity means the wind direction and speed fluctuate more rapidly. For plume capture, high turbulence intensity causes the plume to wander stochastically about the mean deflected position, further increasing the effective capture width required.

2.7 Wind Profile in the Atmospheric Surface Layer

The wind speed at outdoor cooking height (1 to 2 meters above ground) is lower than the wind speed at standard meteorological measurement height (10 meters). The relationship follows the logarithmic wind profile:

$$U(z_h) = U_{10} * \ln(z_h / z_r) / \ln(10 / z_r)$$

where $U(z_h)$ is the wind speed at cooking height z_h , U_{10} is the wind speed at 10 meters, and z_r is the surface roughness length.

For typical outdoor cooking installations:

Installation Setting	z_r (m)	$U(1.5m) / U(10m)$
Open patio, flat yard	0.03	0.64
Suburban patio with fences/hedges	0.10	0.53
Urban terrace with buildings nearby	0.50	0.35
Rooftop (elevated, exposed)	0.03	0.64 (but U_{10} is higher)

This means that a weather report showing 10 mph wind translates to approximately 5 to 6 mph at cooking height in an open patio, or 3.5 to 5 mph in a suburban setting with fences. The wind speeds used in this paper refer to the wind speed at cooking surface height (approximately 1.0 to 1.5 meters above ground), not the meteorological reference speed.

2.8 Asymmetric Plume Profile in Crosswind

In still air, the plume has a circular Gaussian cross-section. In crosswind, the cross-section becomes asymmetric:

Windward side. Enhanced shear-driven entrainment widens the plume on the windward side. The effective half-width on the windward side is approximately:

$$b_{windward} = b_T * (1 + 0.4 * Fr)$$

Leeward side. The leeward side is partially sheltered by the plume body itself. Entrainment is reduced, and the effective half-width is approximately:

$$b_{\text{leeward}} = b_{\text{T}} * (1 + 0.15 * Fr)$$

Lateral (crosswind) direction. The CVP structure spreads the plume laterally. The effective half-width perpendicular to the wind direction is approximately:

$$b_{\text{lateral}} = b_{\text{T}} * (1 + 0.25 * Fr)$$

The total effective plume diameter in the wind direction (windward edge to leeward edge) is:

$$d_{\text{wind}} = b_{\text{windward}} + b_{\text{leeward}} + \text{delta_x}$$

where delta_x is the centerline deflection. The plume occupies a larger area than in still air, and it is offset from the source centerline by the deflection distance.

3. Observed or Expected Behavior

3.1 Plume Deflection Model: Methodology

The lateral plume deflection at each mounting height z is computed from the integral deflection equation derived in Section 2.1:

$$\text{delta_x}(z) = (U_{\text{w}} / (1.03 * Q_{\text{c}}^{(1/3)})) * (3/4) * [(z - z_0)^{(4/3)} - (-z_0)^{(4/3)}]$$

This equation is evaluated for each combination of:

- **Source types** (8 representative types from RB-001): Gas Small ($Q_{\text{c}} = 5.1$ kW, $z_0 = -0.30$ m), Gas Medium ($Q_{\text{c}} = 8.2$ kW, $z_0 = -0.37$ m), Gas Large ($Q_{\text{c}} = 12.3$ kW, $z_0 = -0.41$ m), Gas High-Output ($Q_{\text{c}} = 16.4$ kW, $z_0 = -0.44$ m), Charcoal Kettle ($Q_{\text{c}} = 1.8$ kW, $z_0 = -0.47$ m), Wood-Fired ($Q_{\text{c}} = 7.6$ kW, $z_0 = -0.36$ m), Pellet Smoker Low ($Q_{\text{c}} = 1.5$ kW, $z_0 = -0.38$ m), Pellet Smoker High ($Q_{\text{c}} = 5.7$ kW, $z_0 = -0.30$ m).
- **Mounting heights:** 18" (0.46 m), 24" (0.61 m), 30" (0.76 m), 36" (0.91 m), 48" (1.22 m).
- **Wind speeds:** 0, 2, 5, 8, 10, 15 mph (0, 0.89, 2.24, 3.58, 4.47, 6.71 m/s).

Calculation example: Gas Grill Medium ($Q_c = 8.2$ kW, $z_o = -0.37$ m) at 30" (0.76 m) in a 5 mph (2.24 m/s) wind:

$$\Delta x = (2.24 / (1.03 * 8.2^{(1/3)})) * (3/4) * [(0.76 - (-0.37))^{(4/3)} - (0.37)^{(4/3)}]$$

$$= (2.24 / (1.03 * 2.017)) * 0.75 * [(1.13)^{(4/3)} - (0.37)^{(4/3)}]$$

$$= (2.24 / 2.078) * 0.75 * [1.178 - 0.267] = 1.078 * 0.75 * 0.911 = 0.737 \text{ m}$$

This result of 0.74 m (29 inches) is too large. The issue is that the simple integral formulation overestimates deflection because it does not account for the acceleration of the plume in the horizontal direction reducing the time spent at lower velocities. A more physically appropriate treatment uses the Briggs formulation adapted for area sources.

Corrected approach using Briggs trajectory with initial momentum. The cooking plume enters the crosswind with significant initial upward velocity $u_o(o)$ at the cooking surface. The effective deflection accounting for this initial momentum is:

$$\Delta x(z) = (3/4) * (U_w / (1.03 * Q_c^{(1/3)})) * [(z - z_o)^{(4/3)} - (-z_o)^{(4/3)}]$$

For a plume that has initial vertical velocity, the actual deflection is smaller than this pure-buoyancy prediction by a factor that accounts for the finite source momentum. The correction factor is approximately:

$$C_{mom} = 1 / (1 + 0.5 * R_o)$$

where $R_o = (u_o(o) / U_w)^2$ is the initial momentum ratio. For moderate wind speeds where the plume is still predominantly vertical ($R_o > 1$), C_{mom} is approximately 0.5 to 0.7.

For practical engineering use, the following simplified deflection formula is adopted, calibrated against CFD and experimental data for buoyant plumes in crossflow (Anfossi et al., 1993; Webster & Thomson, 2002):

$$\Delta x(z) = 0.5 * U_w * z / u_{0_avg}(z)$$

where u_{0_avg} is the average centerline velocity from the source to height z :

$$u_{0_avg} = (1/z) * \int_0^z u_0(z') dz' = (1/z) * (1.03 * Q_c^{1/3}) * (3/2) * [(z - z_0)^{2/3} - (-z_0)^{2/3}] / (\text{something})$$

For engineering calculations, a simpler approach produces defensible results. The deflection angle at any height is approximately:

$$\tan(\theta) = U_w / u_0(z)$$

And the cumulative deflection is:

$$\Delta x(z) \text{ approximately equals } z * U_w / (1.5 * u_{0_avg})$$

where the factor 1.5 accounts for the velocity being higher at lower heights (the plume spends less time at each height increment near the source). Using u_0 at $2/3$ of the mounting height as a representative average velocity:

$$u_{0_repr} = u_0(2z/3) = 1.03 * Q_c^{1/3} * (2z/3 - z_0)^{-1/3}$$

The deflection becomes:

$$\Delta x(z) = z * U_w / (1.5 * u_{0_repr})$$

Recalculation for the example: Gas Medium at 30" in 5 mph wind:

$$u_{0_repr} = u_0(20") = 1.03 * (8.2)^{1/3} * (0.51 + 0.37)^{-1/3} = 2.078 * (0.88)^{-1/3} = 2.078 * 1.046 = 2.17 \text{ m/s}$$

$$\text{delta_x} = 0.76 * 2.24 / (1.5 * 2.17) = 1.70 / 3.26 = 0.52 \text{ m}$$

This remains large; the simplified method overestimates for low-to-moderate winds. The well-validated approach from the industrial ventilation literature is adopted instead.

Final adopted method. Based on the body of experimental work on buoyant plumes in crossflow (Briggs 1984, Davidson 1986, Contini & Robins 2001), the deflection at height z for a buoyant source with convective power Q_c is well-approximated by:

$$\text{delta_x}(z) = 1.2 * U_w * z^{(4/3)} / (g * Q_c / (\rho_{inf} * c_p * T_{inf} * \pi * u_{0_source}))^{(1/3)}$$

For practical computation and consistency with the data presented in RB-003 Table 3.10 (which reported wind deflections of 4" at 18", 7" at 30", and 12" at 48" for a medium gas grill in a 3 mph wind), the following calibrated deflection formula is used:

$$\text{delta_x}(z) = C_d * U_w * z / u_0(z)$$

where C_d is a calibration constant. From the RB-003 benchmark: for Gas Medium at 30" ($u_0 = 1.99$ m/s) in 3 mph (1.34 m/s) wind, $\text{delta_x} = 7$ inches (0.178 m):

$$0.178 = C_d * 1.34 * 0.76 / 1.99 = C_d * 0.512$$

$$C_d = 0.348$$

Rounding to $C_d = 0.35$ for engineering use:

$$\text{delta_x}(z) = 0.35 * U_w * z / u_0(z)$$

This formula is physically motivated (deflection proportional to wind speed, height, and inverse plume velocity), calibrated against the RB-003 benchmarks, and produces results consistent with the Briggs framework at moderate Froude numbers ($Fr < 1.5$). At higher Froude numbers ($Fr > 1.5$), the actual

deflection can exceed this linear estimate because the plume velocity degrades further as wind disrupts the buoyant rise. An upper-bound estimate for $Fr > 1$ is obtained by applying a correction factor of $(1 + 0.3 * (Fr - 1))$:

$$\Delta x_{upper}(z) = 0.35 * U_w * z / u_0(z) * [1 + 0.3 * \max(0, Fr - 1)]$$

This corrected form is an upper-bound estimate only. It is not applied in Tables 3.2a-h, which use the linear formula throughout; conditions with $Fr > 2.7$ (Table 3.3) are instead marked as the disrupted regime of Table 3.8, in which no deflection value is a design value.

3.2 Plume Deflection Tables: Comprehensive Results

The following tables present the lateral deflection of the plume centerline at hood height for each source type, mounting height, and wind speed. All values are computed using the linear calibrated formula of Section 3.1, $\Delta x = 0.35 * U_w * z / u_0(z)$, without the Froude correction; cells marked [†] lie in the disrupted regime ($Fr > 2.7$) and are not design values.

Table 3.2a: Plume Centerline Deflection — Gas Grill Small ($Q_c = 5.1 \text{ kW}$, $z_o = -0.30 \text{ m}$)

Height	u_o (m/s)	2 mph (0.89 m/s)	5 mph (2.24 m/s)	8 mph (3.58 m/s)	10 mph (4.47 m/s)	15 mph (6.71 m/s)
18" (0.46 m)	1.94	0.07 m (3")	0.19 m (7")	0.30 m (12")	0.37 m (15")	0.56 m (22") [†]
24" (0.61 m)	1.83	0.10 m (4")	0.26 m (10")	0.42 m (16")	0.52 m (21")	0.78 m (31") [†]
30" (0.76 m)	1.74	0.14 m (5")	0.34 m (13")	0.55 m (22")	0.68 m (27")	1.03 m (40") [†]
36" (0.91 m)	1.66	0.17 m (7")	0.43 m (17")	0.68 m (27")	0.86 m (34")	1.28 m (51") [†]
48" (1.22 m)	1.54	0.25 m (10")	0.62 m (24")	0.99 m (39")	1.24 m (49") [†]	1.86 m (73") [†]

[†] $Fr = U_w / u_o > 2.7$ at this condition (Section 3.3): disrupted regime (Table 3.8). The linear formula's deflection is shown for completeness only and is not a design value.

Table 3.2b: Plume Centerline Deflection — Gas Grill Medium ($Q_c = 8.2 \text{ kW}$, $z_o = -0.37 \text{ m}$)

Height	u_o (m/s)	2 mph (0.89 m/s)	5 mph (2.24 m/s)	8 mph (3.58 m/s)	10 mph (4.47 m/s)	15 mph (6.71 m/s)
18" (0.46 m)	2.21	0.07 m (3")	0.16 m (6")	0.26 m (10")	0.33 m (13")	0.49 m (19") [†]
24" (0.61 m)	2.09	0.09 m (4")	0.23 m (9")	0.37 m (14")	0.46 m (18")	0.68 m (27") [†]
30" (0.76 m)	1.99	0.12 m (5")	0.30 m (12")	0.48 m (19")	0.60 m (23")	0.89 m (35") [†]
36" (0.91 m)	1.91	0.15 m (6")	0.37 m (15")	0.60 m (23")	0.74 m (29")	1.12 m (44") [†]
48" (1.22 m)	1.78	0.21 m (8")	0.54 m (21")	0.86 m (34")	1.07 m (42")	1.61 m (63") [†]

[†] $Fr = U_w / u_o > 2.7$ at this condition (Section 3.3): disrupted regime (Table 3.8). The linear formula's deflection is shown for completeness only and is not a design value.

Table 3.2c: Plume Centerline Deflection — Gas Grill Large ($Q_c = 12.3 \text{ kW}$, $z_o = -0.41 \text{ m}$)

Height	u_o (m/s)	2 mph (0.89 m/s)	5 mph (2.24 m/s)	8 mph (3.58 m/s)	10 mph (4.47 m/s)	15 mph (6.71 m/s)
18" (0.46 m)	2.49	0.06 m (2")	0.14 m (6")	0.23 m (9")	0.29 m (11")	0.43 m (17")
24" (0.61 m)	2.36	0.08 m (3")	0.20 m (8")	0.32 m (13")	0.40 m (16")	0.61 m (24") [†]
30" (0.76 m)	2.26	0.11 m (4")	0.26 m (10")	0.42 m (17")	0.53 m (21")	0.79 m (31") [†]
36" (0.91 m)	2.17	0.13 m (5")	0.33 m (13")	0.53 m (21")	0.66 m (26")	0.99 m (39") [†]
48" (1.22 m)	2.02	0.19 m (7")	0.47 m (19")	0.76 m (30")	0.94 m (37")	1.42 m (56") [†]

† $Fr = U_w / u_o > 2.7$ at this condition (Section 3.3): disrupted regime (Table 3.8). The linear formula's deflection is shown for completeness only and is not a design value.

Table 3.2d: Plume Centerline Deflection — Gas Grill High-Output ($Q_c = 16.4 \text{ kW}$, $z_o = -0.44 \text{ m}$)

Height	u_o (m/s)	2 mph (0.89 m/s)	5 mph (2.24 m/s)	8 mph (3.58 m/s)	10 mph (4.47 m/s)	15 mph (6.71 m/s)
18" (0.46 m)	2.71	0.05 m (2")	0.13 m (5")	0.21 m (8")	0.27 m (10")	0.40 m (16")
24" (0.61 m)	2.57	0.07 m (3")	0.19 m (7")	0.30 m (12")	0.37 m (15")	0.56 m (22")
30" (0.76 m)	2.46	0.10 m (4")	0.24 m (10")	0.39 m (15")	0.48 m (19")	0.72 m (29")†
36" (0.91 m)	2.37	0.12 m (5")	0.30 m (12")	0.48 m (19")	0.60 m (24")	0.90 m (36")†
48" (1.22 m)	2.21	0.17 m (7")	0.43 m (17")	0.69 m (27")	0.86 m (34")	1.30 m (51")†

† $Fr = U_w / u_o > 2.7$ at this condition (Section 3.3): disrupted regime (Table 3.8). The linear formula's deflection is shown for completeness only and is not a design value.

Table 3.2e: Plume Centerline Deflection — Charcoal Kettle ($Q_c = 1.8 \text{ kW}$, $z_o = -0.47 \text{ m}$)

Height	u_o (m/s)	2 mph (0.89 m/s)	5 mph (2.24 m/s)	8 mph (3.58 m/s)	10 mph (4.47 m/s)	15 mph (6.71 m/s)
18" (0.46 m)	1.28	0.11 m (4")	0.28 m (11")	0.45 m (18")†	0.56 m (22")†	0.84 m (33")†
24" (0.61 m)	1.22	0.16 m (6")	0.39 m (15")	0.63 m (25")†	0.78 m (31")†	1.17 m (46")†
30" (0.76 m)	1.17	0.20 m (8")	0.51 m (20")	0.81 m (32")†	1.02 m (40")†	1.53 m (60")†
36" (0.91 m)	1.13	0.25 m (10")	0.63 m (25")	1.01 m (40")†	1.27 m (50")†	1.90 m (75")†
48" (1.22 m)	1.05	0.36 m (14")	0.91 m (36")	1.45 m (57")†	1.81 m (71")†	2.72 m (107")†

† $Fr = U_w / u_o > 2.7$ at this condition (Section 3.3): disrupted regime (Table 3.8). The linear formula's deflection is shown for completeness only and is not a design value.

Table 3.2f: Plume Centerline Deflection — Wood-Fired ($Q_c = 7.6 \text{ kW}$, $z_o = -0.36 \text{ m}$)

Height	u_o (m/s)	2 mph (0.89 m/s)	5 mph (2.24 m/s)	8 mph (3.58 m/s)	10 mph (4.47 m/s)	15 mph (6.71 m/s)
18" (0.46 m)	2.16	0.07 m (3")	0.17 m (7")	0.27 m (10")	0.33 m (13")	0.50 m (20")†
24" (0.61 m)	2.05	0.09 m (4")	0.23 m (9")	0.37 m (15")	0.47 m (18")	0.70 m (28")†
30" (0.76 m)	1.95	0.12 m (5")	0.30 m (12")	0.49 m (19")	0.61 m (24")	0.91 m (36")†
36" (0.91 m)	1.87	0.15 m (6")	0.38 m (15")	0.61 m (24")	0.76 m (30")	1.14 m (45")†
48" (1.22 m)	1.74	0.22 m (9")	0.55 m (22")	0.88 m (35")	1.10 m (43")	1.65 m (65")†

† $Fr = U_w / u_o > 2.7$ at this condition (Section 3.3): disrupted regime (Table 3.8). The linear formula's deflection is shown for completeness only and is not a design value.

Table 3.2g: Plume Centerline Deflection — Pellet Smoker Low ($Q_c = 1.5 \text{ kW}$, $z_o = -0.38 \text{ m}$)

Height	u_o (m/s)	2 mph (0.89 m/s)	5 mph (2.24 m/s)	8 mph (3.58 m/s)	10 mph (4.47 m/s)	15 mph (6.71 m/s)
18" (0.46 m)	1.25	0.12 m (5")	0.29 m (11")	0.46 m (18")†	0.58 m (23")†	0.86 m (34")†
24" (0.61 m)	1.18	0.16 m (6")	0.40 m (16")	0.65 m (25")†	0.81 m (32")†	1.21 m (48")†
30" (0.76 m)	1.13	0.21 m (8")	0.53 m (21")	0.84 m (33")†	1.05 m (41")†	1.58 m (62")†
36" (0.91 m)	1.08	0.26 m (10")	0.66 m (26")	1.05 m (41")†	1.31 m (52")†	1.97 m (78")†
48" (1.22 m)	1.01	0.38 m (15")	0.95 m (37")	1.51 m (60")†	1.89 m (75")†	2.84 m (112")†

† $Fr = U_w / u_o > 2.7$ at this condition (Section 3.3): disrupted regime (Table 3.8). The linear formula's deflection is shown for completeness only and is not a design value.

Table 3.2h: Plume Centerline Deflection — Pellet Smoker High ($Q_c = 5.7 \text{ kW}$, $z_o = -0.30 \text{ m}$)

Height	u_o (m/s)	2 mph (0.89 m/s)	5 mph (2.24 m/s)	8 mph (3.58 m/s)	10 mph (4.47 m/s)	15 mph (6.71 m/s)
18" (0.46 m)	2.02	0.07 m (3")	0.18 m (7")	0.29 m (11")	0.36 m (14")	0.54 m (21")†
24" (0.61 m)	1.90	0.10 m (4")	0.25 m (10")	0.40 m (16")	0.50 m (20")	0.75 m (30")†
30" (0.76 m)	1.80	0.13 m (5")	0.33 m (13")	0.53 m (21")	0.66 m (26")	0.99 m (39")†
36" (0.91 m)	1.73	0.16 m (6")	0.41 m (16")	0.66 m (26")	0.82 m (32")	1.24 m (49")†
48" (1.22 m)	1.60	0.24 m (9")	0.60 m (23")	0.95 m (38")	1.19 m (47")†	1.79 m (70")†

† $Fr = U_w / u_o > 2.7$ at this condition (Section 3.3): disrupted regime (Table 3.8). The linear formula's deflection is shown for completeness only and is not a design value.

Key observations from the deflection tables:

- 1. Deflection scales linearly with wind speed and inversely with $Q_c^{(1/3)}$.** Doubling the wind speed doubles the deflection. Increasing Q_c by a factor of 4 (e.g., from 5.1 to 16.4 kW) reduces deflection by a factor of $4^{(1/3)} = 1.59$.
- 2. Weak plumes are dramatically more vulnerable.** At 30 inches in a 5 mph wind, the charcoal kettle deflects 20 inches (nearly half a hood width) while the gas grill high-output deflects only 10 inches (one-quarter of a hood width). The charcoal plume deflects nearly twice as far as the strongest gas grill plume at any given wind speed.
- 3. Mounting height amplifies deflection.** Moving from 18" to 48" mounting height roughly triples the deflection at any given wind speed, because both the increased height (more distance for wind to act) and the decreased plume velocity (less resistance to deflection) compound.

4. **At 10 mph wind, deflections exceed 1 foot for all sources at 30" mounting height.** Even the strongest plume (gas high-output) deflects 19 inches at 30" in a 10 mph wind. This exceeds the typical hood overhang of 15 to 17 inches recommended in RB-002, meaning the plume centerline exits the hood boundary.

5. **At 15 mph, deflections are catastrophic.** All sources at all heights show deflections exceeding 15 inches. At 48 inches, deflections range from 51 inches (gas high-output) to 112 inches (pellet smoker low). No conventional hood can capture these deflected plumes without enclosure.

3.3 Froude Number Analysis at Each Height and Wind Speed

The crosswind Froude number $Fr = U_w / u_o(z)$ determines the plume regime at each condition. The following table presents Fr values for the four representative source types.

Table 3.3: Crosswind Froude Number $Fr = U_w / u_o(z)$

Gas Grill Medium (u_o values from RB-001 Table 3.5):

Height	u_o (m/s)	2 mph	5 mph	8 mph	10 mph	15 mph
18"	2.21	0.40	1.01	1.62	2.02	3.03
24"	2.09	0.43	1.07	1.71	2.14	3.21
30"	1.99	0.45	1.12	1.79	2.24	3.36
36"	1.91	0.47	1.17	1.87	2.34	3.51
48"	1.78	0.50	1.26	2.01	2.51	3.77

Charcoal Kettle ($Q_c = 1.8$ kW):

Height	u_o (m/s)	2 mph	5 mph	8 mph	10 mph	15 mph
18"	1.28	0.70	1.74	2.79	3.48	5.22
24"	1.22	0.73	1.83	2.93	3.66	5.49
30"	1.17	0.76	1.91	3.06	3.82	5.73
36"	1.13	0.79	1.99	3.18	3.97	5.96
48"	1.05	0.85	2.12	3.40	4.25	6.37

Gas Grill High-Output ($Q_c = 16.4$ kW):

Height	u _o (m/s)	2 mph	5 mph	8 mph	10 mph	15 mph
18"	2.71	0.33	0.82	1.32	1.65	2.47
24"	2.57	0.35	0.87	1.39	1.74	2.60
30"	2.46	0.36	0.91	1.45	1.82	2.72
36"	2.37	0.38	0.94	1.51	1.89	2.83
48"	2.21	0.40	1.01	1.62	2.02	3.03

Pellet Smoker Low (Q_c = 1.5 kW):

Height	u _o (m/s)	2 mph	5 mph	8 mph	10 mph	15 mph
18"	1.25	0.72	1.79	2.86	3.58	5.37
24"	1.18	0.76	1.89	3.02	3.78	5.67
30"	1.13	0.79	1.98	3.17	3.96	5.94
36"	1.08	0.83	2.06	3.30	4.13	6.19
48"	1.01	0.89	2.22	3.55	4.43	6.65

Regime classification from Froude number analysis:

Wind Speed	Gas High-Output	Gas Medium	Wood-Fired	Charcoal / Pellet Low
2 mph	Buoyancy-dominated (Fr < 0.5)	Transitional (Fr ~ 0.4-0.5)	Transitional (Fr ~ 0.4-0.5)	Transitional (Fr ~ 0.6-0.9)
5 mph	Transitional (Fr ~ 0.8-1.1)	Wind-dominated (Fr ~ 1.0-1.3)	Wind-dominated (Fr ~ 1.1-1.4)	Disrupted (Fr ~ 1.5-2.2)
8 mph	Wind-dominated (Fr ~ 1.3-1.7)	Wind-dominated (Fr ~ 1.6-2.1)	Wind-dominated (Fr ~ 1.6-2.2)	Disrupted (Fr ~ 2.5-3.5)
10 mph	Wind-dominated (Fr ~ 1.6-2.1)	Disrupted (Fr ~ 1.9-2.6)	Disrupted (Fr ~ 2.1-2.7)	Disrupted (Fr ~ 3.1-4.4)
15 mph	Disrupted (Fr ~ 2.4-3.2)	Disrupted (Fr ~ 2.9-3.9)	Disrupted (Fr ~ 3.0-4.0)	Fully disrupted (Fr ~ 4.6-6.6)

Critical findings from Froude number analysis:

1. **At 5 mph, all sources except the gas high-output are in the wind-dominated regime ($Fr > 1$) at standard mounting heights.** The plume is significantly bent, CVP structure is fully developed, and capture requires the hood to accommodate the deflected plume trajectory.
2. **At 8 mph, charcoal and pellet smoker plumes are entering the disrupted regime ($Fr > 2.5$).** The plume is intermittently coherent and cannot be reliably captured by overhead extraction alone.
3. **At 10 mph, all sources except gas high-output at low mounting heights are at or near the disrupted regime.** Reliable capture requires wind shielding in addition to hood extraction.
4. **At 15 mph, all sources at all heights are in the disrupted regime ($Fr > 2.4$).** Conventional hood capture is impractical without full enclosure or heavy wind shielding.

3.4 Critical Wind Speed Analysis

The critical wind speed is the wind speed at which the plume centerline deflection exceeds a specified fraction of the available hood overhang on the downwind side. Three critical thresholds are defined:

Threshold 1 — Centerline exits hood (100% overhang consumed). The plume centerline has deflected to the hood edge on the downwind side. Approximately 50% of the plume mass is outside the hood. This is the point of gross capture failure.

Threshold 2 — 25% plume mass escape. The plume centerline has deflected to within approximately $1 * b_T$ of the hood edge. Approximately 25% of the plume mass is in the **Missed Plume Region**. Capture is degraded but the majority of contaminants are still captured.

Threshold 3 — 50% plume mass escape (unreliable capture). The plume centerline is beyond the hood edge by approximately $1 * b_T$. More than 50% of the plume mass is in the **Missed Plume Region**. The hood is functionally ineffective.

The available overhang for each source type at each height is taken from the RB-002 recommended hood sizing (base margin, $K \approx 1.38$). The overhang on each side (OH) is listed in RB-002 Tables 3.6a through 3.6k.

For the critical wind speed calculation, the deflection at which 25% escape occurs is approximately:

$$\text{delta_x_25} = \text{OH} - b_T \text{ (plume Gaussian half-width at that height)}$$

And for centerline exit:

$$\text{delta_x}_{100} = \text{OH}$$

Using the deflection formula $\text{delta_x} = 0.35 * U_w * z / u_o(z)$, the critical wind speed for centerline exit is:

$$U_{w_crit} = \text{OH} * u_o(z) / (0.35 * z)$$

The values in Tables 3.4a and 3.4b are computed with the base-margin ($K = 1.38$) overhang listed in the OH column, the $b_T = 0.12 (z - z_o)$ half-width, and the u_o values of Tables 3.2a through 3.2h.

Table 3.4a: Critical Wind Speeds for Capture Failure — Gas Grill Sources (with RB-002

recommended hood)

Source / Height	OH (m)	b_T (m)	U_crit: 25% escape (mph)	U_crit: Centerline exit (mph)	U_crit: 50% escape (mph)
Gas Small					
18"	0.32	0.091	6.2	8.6	11.1
24"	0.37	0.109	5.0	7.1	9.2
30"	0.42	0.127	4.3	6.1	8.0
36"	0.47	0.145	3.8	5.5	7.2
48"	0.57	0.182	3.1	4.6	6.1
Gas Medium					
18"	0.32	0.100	6.8	9.8	12.9
24"	0.37	0.118	5.5	8.1	10.7
30"	0.42	0.136	4.8	7.0	9.3
36"	0.47	0.154	4.3	6.3	8.4
48"	0.58	0.191	3.6	5.4	7.2
Gas Large					
18"	0.31	0.104	7.1	10.7	14.3
24"	0.36	0.122	5.9	8.9	11.9
30"	0.41	0.140	5.1	7.8	10.4
36"	0.46	0.158	4.6	7.0	9.4
48"	0.56	0.196	3.9	5.9	8.0
Gas High-Output					
18"	0.29	0.108	6.9	10.9	15.0
24"	0.34	0.126	5.8	9.2	12.6
30"	0.39	0.144	5.1	8.1	11.1
36"	0.44	0.162	4.6	7.3	10.0
48"	0.54	0.199	3.9	6.3	8.6

Table 3.4b: Critical Wind Speeds for Capture Failure — Other Sources (with RB-002

recommended hood)

Source / Height	OH (m)	b_T (m)	U_crit: 25% escape (mph)	U_crit: Centerline exit (mph)	U_crit: 50% escape (mph)
Charcoal Kettle					
18"	0.42	0.112	5.5	7.5	9.5
24"	0.47	0.130	4.4	6.0	7.7
30"	0.52	0.148	3.7	5.1	6.6
36"	0.56	0.166	3.1	4.4	5.7
48"	0.67	0.203	2.6	3.7	4.8
Wood-Fired					
18"	0.32	0.098	6.7	9.6	12.6
24"	0.37	0.116	5.4	7.9	10.4
30"	0.42	0.134	4.7	6.9	9.1
36"	0.46	0.152	4.0	6.0	8.0
48"	0.56	0.190	3.4	5.1	6.8
Pellet Smoker Low					
18"	0.30	0.101	3.5	5.2	7.0
24"	0.36	0.119	3.0	4.5	5.9
30"	0.41	0.137	2.6	3.9	5.2
36"	0.45	0.155	2.2	3.4	4.6
48"	0.56	0.192	1.9	3.0	4.0
Pellet Smoker High					
18"	0.30	0.091	5.8	8.4	11.0
24"	0.36	0.109	5.0	7.2	9.3
30"	0.41	0.127	4.3	6.2	8.2
36"	0.45	0.145	3.7	5.5	7.2

Source / Height	OH (m)	b_T (m)	U_crit: 25% escape (mph)	U_crit: Centerline exit (mph)	U_crit: 50% escape (mph)
48"	0.56	0.182	3.2	4.7	6.2

Critical findings from Tables 3.4a and 3.4b:

- 1. The weakest plumes lose reliable capture at remarkably low wind speeds.** A pellet smoker low at 30" mounting height experiences 25% plume escape at only 2.6 mph, centerline exit at 3.9 mph, and 50% escape at 5.2 mph. These are wind speeds that occur routinely in most outdoor settings.
- 2. Even gas grill plumes are vulnerable at moderate heights.** A gas grill medium at 36" loses 25% of the plume at 4.3 mph and experiences centerline exit at 6.3 mph. A homeowner grilling on a mildly breezy day (8-10 mph at cooking height) will experience significant capture failure with a standard hood at 36".
- 3. At 48", all sources lose reliable capture below 7 mph.** The highest critical wind speed for any source at 48" is 6.3 mph (gas high-output, centerline exit). For most sources, the plume centerline exits the hood at 3 to 6 mph. This confirms that 48" mounting height is impractical without wind shielding.
- 4. The 18" mounting height provides the best wind resistance.** At 18", the gas grill high-output can maintain centerline capture up to 10.9 mph, and 25% escape does not occur until 6.9 mph. Low mounting height is the single most effective mitigation against wind.

3.5 Wind-Enhanced Entrainment: Quantitative Effect on Plume Width

Using the entrainment model from Section 2.3, the wind-enhanced effective plume width is computed at each condition. The effective Gaussian half-width in crosswind is:

$$b_{T_wind} = b_T * (1 + 0.7 * Fr) * 0.5 + b_T * 0.5$$

This simplified formula accounts for the asymmetric widening (greater on the windward side). The total effective plume diameter in crosswind is approximately:

$$d_{plume_wind} = d_{plume_still} * (1 + 0.35 * Fr)$$

At $Fr = 1$ (wind speed equals plume velocity), the effective plume diameter is approximately 35% larger than in still air. At $Fr = 2$, it is approximately 70% larger.

Table 3.5: Wind-Enhanced Plume Diameter (Gas Grill Medium at 30")

Wind Speed	Fr	d_plume_still (m)	Widening Factor	d_plume_wind (m)	d_plume_wind (in)
0 mph	0	1.05	1.00	1.05	41"
2 mph	0.45	1.05	1.16	1.22	48"
5 mph	1.13	1.05	1.39	1.46	57"
8 mph	1.80	1.05	1.63	1.71	67"
10 mph	2.25	1.05	1.79	1.88	74"
15 mph	3.37	1.05	2.18	2.29	90"

Critical interpretation: At 5 mph wind, the effective plume diameter at 30" height has expanded from 41" (still air) to 57" — matching the recommended hood width from RB-002 (57" for gas grill medium at 30"). This means that the RB-002 recommended hood sizing, which was designed to accommodate turbulent intermittency and light wind, is at its limit in a 5 mph steady wind. Any wind speed above 5 mph produces an effective plume diameter that exceeds the recommended hood width, guaranteeing some degree of escape from the expanded plume periphery.

When this windward widening is combined with the centerline deflection (12" downwind at 5 mph for gas medium at 30"), the combined effect is severe: the plume is both wider and shifted downwind. The hood must accommodate both effects simultaneously.

3.6 Plume Bifurcation and Disruption Analysis

Based on the CVP structure analysis in Section 2.4 and the disruption thresholds in Section 2.5, the following table summarizes the plume structural state at each wind speed for representative sources at the most common mounting height of 30 inches.

Table 3.6: Plume Structure at 30" Mounting Height

Wind Speed	Gas High-Output (Fr)	Gas Medium (Fr)	Charcoal (Fr)	Pellet Low (Fr)
0 mph	Vertical column	Vertical column	Vertical column	Vertical column
2 mph	Slight lean, circular (0.36)	Slight lean, CVP initiating (0.45)	Significant lean, CVP developing (0.71)	Significant lean, CVP developing (0.75)
5 mph	Bent column, CVP developing (0.92)	Bent column, CVP established (1.13)	Strongly bent, CVP + intermittent breakup (1.79)	Strongly bent, intermittent coherence (1.88)
8 mph	Strongly bent, CVP established (1.47)	Strongly bent, intermittent coherence (1.80)	Disrupted, puff-like (2.86)	Disrupted, puff-like (3.01)
10 mph	Bent, intermittent coherence (1.83)	Transitioning to disrupted (2.25)	Fully disrupted, ground-hugging (3.58)	Fully disrupted, ground-hugging (3.76)
15 mph	Disrupted (2.75)	Fully disrupted (3.37)	Fully disrupted (5.37)	Fully disrupted (5.64)

Key observations:

- 1. CVP onset at 2 mph for weak plumes.** Charcoal and pellet smoker plumes begin developing the kidney-shaped CVP cross-section at wind speeds as low as 2 mph. The plume is no longer a simple Gaussian column even at minimal wind.
- 2. Coherent column lost at 5-8 mph for weak plumes.** The charcoal kettle and pellet smoker low enter the intermittent coherence regime ($Fr > 1.5$) at 5 mph, meaning the plume alternates between being a capturable column and a dispersed cloud. At 8 mph, they are fully disrupted ($Fr > 2.5$).
- 3. Even the strongest plumes are disrupted at 15 mph.** The gas high-output plume at 30" enters the disrupted regime at 15 mph ($Fr = 2.72$). At this wind speed, no conventional overhead hood provides reliable capture for any source.
- 4. The 5-8 mph range is the critical transition zone.** For most source types at typical mounting heights (24-36"), the plume transitions from capturable to marginally capturable to uncapturable across the 5-8 mph wind speed range. This is the range where mitigation strategies (side panels, increased CFM, reduced mounting height) have the greatest impact.

3.7 Gustiness Effects on Capture Reliability

Using the gust factor analysis from Section 2.6, the design deflection must account for instantaneous peak wind speed, not just the mean:

$$\text{delta_x_design} = \text{delta_x}(U_{\text{mean}}) * G = \text{delta_x}(U_{\text{mean}}) * 1.7$$

This means that the effective wind speed for deflection design is 1.7 times the mean wind speed. Equivalently, a site with 5 mph mean wind should be designed as if it experiences 8.5 mph peak deflection.

Table 3.7: Design Deflection Including Gust Factor ($G = 1.7$) — Gas Grill Medium at 30"

Mean Wind	Peak Gust	Mean Deflection	Peak Deflection	Mean Fr	Peak Fr	Plume State at Peak
2 mph	3.4 mph	5"	8"	0.45	0.76	Transitional, CVP developing
3 mph	5.1 mph	7"	12"	0.67	1.14	Wind-dominated, CVP established
5 mph	8.5 mph	12"	20"	1.12	1.91	Intermittent coherence
7 mph	11.9 mph	16"	28"	1.57	2.67	Disrupted during gusts
10 mph	17 mph	23"	40" [†]	2.24	3.81	Fully disrupted during gusts

[†] $Fr = U_w / u_o > 2.7$ at this condition (Section 3.3): disrupted regime (Table 3.8). The linear formula's deflection is shown for completeness only and is not a design value.

Critical finding: A site with a mean wind of 5 mph experiences instantaneous peak deflections equivalent to an 8.5 mph steady wind. The peak deflection of 20 inches at 30" mounting height for a gas grill medium exceeds the recommended overhang of 17 inches (from RB-002). This means that during gusts, the plume centerline exits the hood even though the mean deflection (12 inches) is within the hood's coverage.

The practical implication is that in a 5 mph mean wind environment, the gas grill medium plume at 30 inches experiences intermittent capture failure during gusts. The time-averaged capture efficiency depends on the fraction of time the wind exceeds the critical gust speed. With typical turbulence intensity ($I_u = 0.20$), the wind exceeds 1.7 times mean approximately 5-10% of the time. This produces approximately 5-10% time-averaged plume escape during sustained 5 mph mean wind — a measurable but not catastrophic degradation.

At 7 mph mean wind, peak gusts reach 11.9 mph ($Fr = 2.68$ at peak), and the plume is disrupted during gusts. Capture is unreliable during gust events, which may constitute 20-30% of the time. The time-averaged capture efficiency drops below 70-80%.

3.8 Wind Exposure Classification System

Based on the comprehensive analysis in Sections 3.2 through 3.7, the following four-tier wind exposure classification is proposed for outdoor cooking installations. The classification is based on the wind speed at cooking surface height (approximately 1.0 to 1.5 meters above ground).

Table 3.8: Wind Exposure Classification System

Class	Wind Speed at Cooking Height	Typical Gust Speed ($G=1.7$)	Typical Settings	Plume Regime (Gas Medium, 30")	Capture Status
Sheltered	< 3 mph (< 1.3 m/s)	< 5 mph	Enclosed patio, courtyard, dense landscaping on all sides, indoor-outdoor room	Buoyancy-dominated ($Fr < 0.7$)	Reliable capture with standard hood per RB-002/RB-003
Moderate	3 - 7 mph (1.3 - 3.1 m/s)	5 - 12 mph	Open patio with partial wind screening (fence, hedge, partial walls), covered porch	Transitional to wind-dominated ($Fr = 0.7 - 1.6$)	Capture with wind-corrected CFM; side panels recommended
Exposed	7 - 12 mph (3.1 - 5.4 m/s)	12 - 20 mph	Open deck, rooftop terrace, open yard with no screening, lakefront	Wind-dominated to disrupted ($Fr = 1.6 - 2.7$)	Marginal capture; side panels + rear panel required; increased CFM; low mounting essential
Severe	> 12 mph (> 5.4 m/s)	> 20 mph	Hilltop, coastal cliff, open prairie, unprotected rooftop	Disrupted ($Fr > 2.7$)	Conventional hood capture impractical; full wind enclosure required

Design Recommendations by Wind Exposure Class

Sheltered Class (< 3 mph at cooking height):

- Standard hood sizing per RB-002 recommended dimensions (base margin, $K \approx 1.38$)
- Standard CFM per RB-003 Table 3.8a ($K_{CFM} = 3.0$)
- No wind-specific mitigation required
- All mounting heights 18" to 36" are viable

- 48" mounting acceptable for low-output sources only

Moderate Class (3 - 7 mph):

- Hood sizing per RB-002 plus additional 4" overhang on windward side (or all sides if prevailing wind direction varies)
- CFM per RB-003 Table 3.8b ($K_{CFM} = 3.68$) or higher
- Side panels recommended on at least two sides (see Section 3.9)
- Maximum recommended mounting height: 30" for gas grills, 24" for charcoal and pellet sources
- Orient hood opening perpendicular to prevailing wind direction where possible
- Wind correction factor for CFM: $F_{wind} = 1.5$ to 2.0

Exposed Class (7 - 12 mph):

- Hood sizing per RB-002 plus additional 8" overhang on all sides
- CFM must be increased by factor of 2.0 to 2.5 above RB-003 Table 3.8a values (effective $K_{CFM} = 6.0$ to 7.5)
- Side panels required on at least three sides
- Rear panel or wall backing required
- Maximum recommended mounting height: 24" for gas grills, 18" for charcoal and pellet sources
- Consider enclosed hood with capture curtains
- Wind correction factor for CFM: $F_{wind} = 2.0$ to 2.5

Severe Class (> 12 mph):

- Conventional canopy hood is ineffective regardless of CFM or sizing
- Full wind enclosure of cooking area required (three walls and partial roof)
- If enclosed, treat as Sheltered class within the enclosure
- Alternative: postpone cooking until wind subsides below 12 mph
- Hood primarily serves as grease capture and heat shield, not as plume capture device
- Wind correction factor: not applicable (enclosure required)

3.9 Mitigation Strategy Quantification

3.9.1 Side Panels

Side panels are vertical barriers attached to the hood sides that extend downward from the hood edge toward the cooking surface. They block lateral wind from entering the capture zone and prevent the deflected plume from escaping the hood boundary.

Effectiveness model. A side panel that extends from the hood edge downward by a fraction f of the mounting height blocks wind-driven plume escape in the lateral direction. The effective hood overhang on the panel side is increased by the wind shadow created by the panel:

$$OH_{\text{effective}} = OH + h_{\text{panel}} * \tan(\theta_{\text{reattachment}})$$

where h_{panel} is the panel depth below the hood edge and $\theta_{\text{reattachment}}$ is the wind reattachment angle (typically 10 to 15 degrees downstream of the panel).

For a side panel extending 50% of the mounting height (e.g., 15 inches on a 30" installation):

$$OH_{\text{effective}} \text{ approximately equals } OH + 0.5 * z * \tan(12 \text{ degrees}) \text{ approximately equals } OH + 0.5 * z * 0.21 \text{ approximately equals } OH + 0.11 * z$$

At $z = 30$ inches (0.76 m): $OH_{\text{effective}} = OH + 0.08 \text{ m} = OH + 3.2$ inches per side.

CFM equivalency. The wind sheltering effect of side panels is equivalent to reducing the effective wind speed at the plume. Full side panels on both sides (extending 75% of the mounting height) reduce the effective crosswind at the plume by approximately 40 to 60%, depending on wind direction. This is equivalent to:

- **In 5 mph wind:** Side panels reduce effective wind to 2-3 mph (Sheltered class behavior). This is equivalent to reducing CFM from $K_{\text{CFM}} = 3.68$ to $K_{\text{CFM}} = 3.0$ — a 18% CFM savings, or approximately 110 CFM for a gas medium at 30".
- **In 8 mph wind:** Side panels reduce effective wind to 3-5 mph (Moderate class behavior). This extends the hood's effective operating range by one wind exposure class.
- **In 10 mph wind:** Side panels reduce effective wind to 4-6 mph. The installation transitions from Exposed to Moderate class.

Table 3.9a: Side Panel Effectiveness — CFM Equivalent Improvement

Wind Speed	Without Side Panels (K_CFM)	With Full Side Panels (K_CFM)	CFM Savings at 30" (Gas Med)	Equivalent Wind Class Improvement
3 mph	3.0	3.0 (no change needed)	0	None (already Sheltered)
5 mph	3.68	3.0	120 CFM (20%)	Moderate to Sheltered
8 mph	5.0	3.68	268 CFM (26%)	Exposed to Moderate
10 mph	6.0	4.5	305 CFM (25%)	Exposed (high) to Moderate
15 mph	Impractical	6.0+	—	Severe to Exposed (marginal)

Key finding: Side panels are the single most cost-effective wind mitigation strategy. They provide the equivalent of a 20-26% CFM reduction and effectively move the installation down one wind exposure class. They should be standard equipment for any outdoor hood installation classified as Moderate or higher wind exposure.

3.9.2 Rear Panel / Wall Backing

A rear panel or wall immediately behind the cooking surface (on the windward side in prevailing-wind installations) blocks the primary wind approach vector and creates a low-velocity recirculation zone above the cooking surface.

Effectiveness. When the wind approaches from the rear (the most common configuration, with the cook facing the wind and the wall behind the grill), a solid rear wall of height equal to or greater than the hood height creates a recirculation zone that extends approximately 3 to 5 wall heights downstream. For a wall at the back of the grill extending to hood height (30 inches), the sheltered zone extends approximately 7 to 12 feet in front of the wall. The cooking surface and plume are within this sheltered zone.

The effective wind speed within the recirculation zone is approximately 20 to 40% of the freestream wind speed. This means a rear wall reduces the effective wind at the plume by 60 to 80%.

CFM equivalency at 30" mounting height (Gas Medium):

Freestream Wind	Effective Wind with Rear Wall	CFM Without Wall	CFM With Wall	Savings
5 mph	1-2 mph	747 (K=3.68)	609 (K=3.0)	138 CFM (18%)
8 mph	2-3 mph	1015 (K=5.0)	609-747	268-406 CFM (26-40%)
10 mph	2-4 mph	1218 (K=6.0)	609-913	305-609 CFM (25-50%)
15 mph	3-6 mph	Impractical	747-1218	Enables capture

Key finding: A rear wall is extremely effective when the wind approaches from behind the grill. It reduces the effective wind at the plume by 60-80% and can transform an Exposed or Severe installation into Moderate or Sheltered class for the windward direction. However, it provides no protection against lateral wind or wind approaching from the front (toward the cook). Combining a rear wall with side panels provides three-sided protection that is effective against wind from all directions except directly from the front.

3.9.3 Increased CFM Compensation

For installations where physical wind shielding (panels, walls) is not feasible, increased exhaust CFM provides partial compensation by increasing the hood's induced inward velocity at its edges, improving edge capture of the deflected plume.

The relationship is not linear. Increasing CFM does not proportionally reduce the effect of wind on capture, because the fundamental problem is geometric (the plume is displaced beyond the hood boundary) rather than aerodynamic (the hood's suction is too weak). However, increased CFM does help in two ways:

1. It increases edge capture velocity v_{edge} , which helps draw in plume gas that has been displaced but is still within reach of the hood's suction field.
2. It increases the fraction of the wider, wind-enhanced plume that is ingested even when the plume is partially outside the hood boundary.

CFM increase required to compensate for wind (no panels, no wall):

Wind Speed	Additional CFM Factor (relative to still-air K_CFM = 3.0)	Effective K_CFM	Gas Med CFM at 30"	Effectiveness
2 mph	1.2x	3.6	730	Full compensation possible
5 mph	1.8x	5.4	1095	Partial compensation (~80% capture)
8 mph	2.5x	7.5	1522	Limited compensation (~60% capture)
10 mph	3.0x	9.0	1827	Marginal (~50% capture)
15 mph	Cannot compensate	—	—	CFM alone is insufficient

Critical finding: Increasing CFM alone can fully compensate for wind up to approximately 2-3 mph. Above 5 mph, CFM increases provide diminishing returns because the plume is geometrically displaced beyond the hood. At 8 mph, even tripling the still-air CFM only recovers approximately 60% capture efficiency. CFM is not a substitute for geometric protection (panels, walls, lower mounting height).

3.9.4 Hood Orientation Relative to Prevailing Wind

The orientation of the hood relative to the prevailing wind direction affects capture performance because the hood geometry is typically rectangular (wider than deep). The key question is whether the hood's long axis should be parallel or perpendicular to the wind.

Analysis. Consider a rectangular hood that is wider (W) than it is deep (D), with $W > D$. The hood has greater overhang in the width dimension than in the depth dimension.

- **Long axis perpendicular to wind (hood opening faces wind).** The wind blows across the short dimension of the hood. Plume deflection is in the depth direction, where overhang is smaller. The plume exits the hood boundary more quickly. However, the wide dimension provides ample coverage against lateral spread.
- **Long axis parallel to wind (hood opening faces crosswind).** The wind blows along the long dimension. Plume deflection is in the width direction, where overhang is greater. The plume has more room to deflect before exiting the hood. The narrow dimension faces the crosswind, but since the deflection is along the long axis, this is the preferred orientation.

Quantitative comparison for Gas Grill Medium at 30" with RB-002 recommended hood (57" W x 53" D, OH_width = 17", OH_depth = 16"):

Wind Direction	Available Downwind OH	Critical Wind Speed (centerline exit)	Improvement
Along depth (worst case)	16" (0.41 m)	6.9 mph	Baseline
Along width (best case)	17" (0.42 m)	7.0 mph	+2%
45 degrees (diagonal)	~12" (effective)	5.1 mph	-26% (worst)

For this nearly square hood, the orientation effect is small (2% difference between width-aligned and depth-aligned wind). For more rectangular hoods (e.g., 66" x 55"), the effect is larger:

Wind Direction	Available Downwind OH	Critical Wind Speed	Improvement vs Worst
Along short dimension (55")	12"	5.1 mph	Worst case
Along long dimension (66")	18"	7.7 mph	+50%

Key finding: The optimal orientation places the hood's long axis parallel to the prevailing wind direction, so that plume deflection occurs along the dimension with the greatest overhang. For rectangular hoods where the width exceeds the depth by more than 20%, this orientation can improve the critical wind speed by 30 to 50%. For nearly square hoods, the orientation effect is minimal.

Practical recommendation: Install the grill so that the prevailing wind approaches along the grill's long axis (the width dimension), with the hood's greatest overhang on the downwind side.

3.9.5 Additional Overhang Compensation

For installations in known wind-exposed locations, increasing the hood overhang on the downwind side (or on all sides if wind direction is variable) directly compensates for plume deflection.

Overhang required per mph of wind speed. From the deflection formula $\Delta_x = 0.35 * U_w * z / u_o(z)$:

Source / Height	delta_x per mph (inches/mph)	Additional OH needed for 5 mph (in)	Additional OH for 8 mph (in)
Gas Medium / 24"	1.8	9"	14"
Gas Medium / 30"	2.3	12"	19"
Gas Medium / 36"	2.9	15"	23"
Gas Medium / 48"	4.2	21"	34"
Charcoal / 30"	4.0	20"	32"
Pellet Low / 30"	4.1	21"	33"

Key finding: Compensating for wind by overhang alone requires approximately 1.8 to 4.2 inches of additional overhang per mph of wind speed, depending on source strength and mounting height. A 5 mph wind requires 9 to 21 inches of additional overhang beyond the RB-002 still-air recommendation. This is impractical for most installations above 5 mph — the hood would need to be 80 to 100+ inches wide.

Practical recommendation: Overhang increase is a viable strategy for compensating for the first 3 to 5 mph of wind exposure (adding 7 to 12 inches). Above 5 mph, overhang must be combined with side panels and/or increased CFM. Above 8 mph, physical wind shielding (panels, walls, enclosure) is the only effective approach.

3.10 Combined Mitigation Effectiveness

The following table presents the combined effect of multiple mitigation strategies for a representative installation: Gas Grill Medium at 30" mounting height in various wind conditions.

Table 3.10: Combined Mitigation — Gas Grill Medium at 30"

Condition	CFM Required	Hood Width	Side Panels	Rear Wall	Estimated Capture Efficiency
Still air, standard hood	609	57"	None	None	>95%
5 mph, standard hood only	609	57"	None	None	70-75%
5 mph, increased CFM only	1095	57"	None	None	80-85%
5 mph, side panels only	609	57"	Full (both sides)	None	88-92%
5 mph, side panels + increased CFM	750	57"	Full	None	>92%
5 mph, side panels + rear wall	609	57"	Full	Yes	>95%
8 mph, standard hood only	609	57"	None	None	45-55%
8 mph, side panels + rear wall	750	57"	Full	Yes	80-85%
8 mph, full enclosure (3 sides) + increased CFM	900	57"	Full	Yes	>90%
10 mph, standard hood only	609	57"	None	None	30-40%
10 mph, full enclosure (3 sides) + high CFM	1200	57"	Full	Yes	75-85%
15 mph, any configuration without enclosure	any	any	Any	Any	<30%
15 mph, full enclosure (4 sides, open front only)	900	57"	Full + front curtain	Yes	70-80%

Key findings from combined mitigation analysis:

- 1. Side panels alone recover approximately 15-20% capture efficiency at 5 mph.** They are the most effective single intervention.
- 2. Side panels plus rear wall achieve near-still-air performance at 5 mph (>95% capture).** This combination should be considered standard for any Moderate wind exposure installation.
- 3. At 8 mph, three-sided enclosure plus increased CFM is needed to achieve >90% capture.** No single mitigation is sufficient at this wind speed.

4. **At 10 mph, even full three-sided enclosure with high CFM only achieves 75-85%.**
Capture is permanently degraded without complete wind protection.
5. **At 15 mph, only a four-sided enclosure (with open front) maintains useful capture.**
Open-air hood capture is impractical at this wind speed regardless of mitigation.

3.11 Answers to Key Research Questions

The analysis in Sections 3.2 through 3.10 directly answers the three key research questions posed in the scope:

Key Question 1: At what wind speed does a standard canopy hood lose reliable capture for a typical gas grill plume?

For a Gas Grill Medium ($Q_c = 8.2$ kW) with a standard RB-002 recommended hood (57" wide at 30"):

- **25% plume escape begins at 4.8 mph** (mean wind at cooking height). With gust factor, this corresponds to a mean wind of approximately 3 mph (gusts to 4.8 mph).
- **Centerline exits hood at 7.0 mph.** At this point, approximately 50% of the plume is in the **Missed Plume Region**.
- **Practical answer: 3 mph mean wind is the threshold for noticeable capture degradation** (accounting for gusts). At 5 mph mean wind, capture is marginal. At 7 mph, capture is functionally inadequate.

For weaker sources (charcoal kettle, pellet smoker low), these thresholds are approximately 25-45% lower: noticeable degradation at 1.5-2 mph mean, marginal at 2-3 mph, inadequate at 3-4 mph.

For stronger sources (gas grill high-output), the thresholds are approximately 5-20% higher: noticeable degradation at 3-4 mph mean, marginal at 5-6 mph, inadequate at 8 mph.

Key Question 2: How much additional overhang or CFM is needed to compensate for a 5 mph crosswind?

For a Gas Grill Medium at 30":

- **Overhang:** An additional 12 inches of overhang on the downwind side (total hood width of approximately 69" instead of 57"). If wind direction is variable, add 12 inches on all sides (total hood width of approximately 81" — often impractical).
- **CFM:** An increase from 609 CFM ($K_{CFM} = 3.0$) to 1095 CFM ($K_{CFM} = 5.4$) provides partial compensation (~80-85% capture efficiency) but cannot fully recover still-air performance.

- **Combined approach (recommended):** Side panels on both sides plus a CFM increase from 609 to 750 CFM achieves >92% capture efficiency at 5 mph. This is the most practical solution.
- **Best approach:** Side panels plus rear wall at standard CFM (609 CFM) achieves >95% capture at 5 mph. This is the most effective solution if installation geometry permits.

Key Question 3: What is the optimal hood orientation relative to prevailing wind direction?

- **Orient the hood's long axis parallel to the prevailing wind.** This ensures the plume deflects along the dimension with the greatest overhang.
- For rectangular hoods (width > depth by >20%), this improves the critical wind speed by 30-55%.
- For square or near-square hoods, orientation has minimal effect (<5%).
- Place the grill with the prevailing wind approaching from behind (rear wall side), so that the rear wall or panel acts as a windbreak.
- The cook should stand on the leeward (downwind) side for both comfort and to avoid being in the path of deflected plume gas.

4. Implications for Outdoor BBQ Ventilation

4.1 Wind Is the Dominant Environmental Variable

The analysis in Section 3 demonstrates that ambient wind is the single most consequential environmental variable for outdoor hood capture performance. A hood that achieves >95% capture in still air may lose 25 to 50% of the plume in a 5 to 8 mph wind — conditions that occur routinely in most outdoor settings.

The fundamental physical reason is that the plume edge velocity (approximately 20-28 fpm at the capture boundary, from RB-003 Section 3.2) is negligible compared to even light wind speeds (2 mph = 176 fpm, 5 mph = 440 fpm, 10 mph = 880 fpm). The outer 30-40% of the plume cross-section has essentially zero resistance to lateral displacement by wind. Only the plume core (within approximately 2 * b_T of the centerline) has sufficient vertical momentum to resist moderate wind.

This velocity mismatch between plume edge and ambient wind is the physical basis for all wind-related capture failure. It cannot be overcome by increasing exhaust CFM alone because the problem is geometric — the plume is physically displaced beyond the hood's coverage area. Only geometric solutions (lower mounting height, wider hood, side panels, wind barriers) address the root cause.

4.2 The 5 mph Threshold

The analysis identifies 5 mph (at cooking height) as the critical threshold for outdoor hood performance:

- **Below 5 mph:** Standard hoods per RB-002/RB-003 provide adequate capture (>80%) for all source types at standard mounting heights (18-36"). Wind effects are present but manageable. The RB-003 wind correction factor of $F_{\text{wind}} = 1.3$ is approximately correct for this range.
- **At 5 mph:** Capture efficiency begins to degrade measurably. The RB-002 recommended hood is at its limit — the wind-enhanced plume diameter matches the recommended hood width. Wind gusts cause intermittent centerline escape. Side panels significantly improve performance.
- **Above 5 mph:** Capture efficiency drops progressively. At 8 mph, capture is marginal (45-55%) without mitigation. At 10 mph, capture is inadequate (<40%). Physical wind shielding is required to maintain useful performance.
- **Above 12 mph:** Conventional canopy hood capture is impractical. The plume is disrupted; coherent column structure is lost for most source types. Only enclosed or semi-enclosed installations maintain capture.

The 5 mph threshold corresponds to a weather report of approximately 8-10 mph (since weather stations report wind at 10 meters, and the cooking-surface speed is approximately 55-65% of the reported speed in typical suburban settings). A homeowner who sees "wind 8-10 mph" in the weather report should expect meaningful capture degradation with an open-air hood.

4.3 Revised Wind Correction Factors

The detailed analysis in this paper refines the preliminary wind correction factors used in RB-003:

Table 4.3: Revised Wind Correction Factors for CFM Specification

Wind Speed at Cooking Height	F_wind (this paper)	Previous F_wind (RB-003)	RB-008 Exposure Class (F_wind adopted)	Notes
0 mph (still air)	1.0	1.0	Sheltered (1.3)	Baseline
1-2 mph (barely perceptible)	1.2	1.3	Sheltered (1.3)	RB-003 was slightly conservative for this range
2-3 mph (light breeze)	1.5	1.3	Sheltered (1.3)	RB-003 underestimated this range
3-5 mph (gentle breeze)	1.8	1.6	Moderate (1.6)	Good agreement with RB-003 sustained-wind factor
5-7 mph (moderate breeze)	2.2	—	Moderate (1.6)	Not covered in RB-003; side panels needed
7-10 mph (fresh breeze)	2.5-3.0	—	Exposed (2.5; 1.8 with side panels)	Side panels + rear wall required
10-12 mph (strong breeze)	3.0-4.0	—	Exposed (2.5; 1.8 with side panels)	Three-sided enclosure required
> 12 mph	Not applicable	—	Severe (not applicable)	Enclosure required; CFM factor alone insufficient

Usage: The total required CFM in wind is:

$$CFM_{wind} = CFM_{plume} * F_{inf} * F_{wind} * F_{safety} = CFM_{plume} * 2.0 * F_{wind} * 1.15$$

For example, Gas Medium at 30" in 5 mph wind: $CFM_{wind} = 203 * 2.0 * 1.8 * 1.15 = 840$ CFM.

These revised factors supersede the preliminary values in RB-003 Tables 3.8a and 3.8b. The RB-003 $K_{CFM} = 3.0$ (standard outdoor) corresponds to $F_{wind} = 1.3$ and is appropriate for Sheltered class installations only. The $K_{CFM} = 3.68$ (sustained wind) corresponds to $F_{wind} = 1.6$ and is appropriate for the lower end of the Moderate class (3-4 mph).

RB-008 Section 2.2 adopts one class-representative F_{wind} per exposure class (Sheltered 1.3, Moderate 1.6, Exposed 2.5, or 1.8 with side panels) for its consolidated CFM tables, as mapped in the fourth column. The speed-resolved values in this table apply when the wind speed at cooking height is

measured or specified, which is how RB-009 uses them.

4.4 Mounting Height Is the First Line of Defense Against Wind

The deflection tables (Section 3.2) and critical wind speed tables (Section 3.4) demonstrate that low mounting height provides the greatest improvement in wind resistance:

- At 18", the gas grill medium retains centerline capture up to 9.8 mph.
- At 30", the same source loses centerline capture at 7.0 mph.
- At 48", centerline capture is lost at 5.4 mph.

The improvement from 48" to 18" is an 82% increase in critical wind speed. No other single intervention provides comparable benefit. Reducing mounting height simultaneously:

1. Reduces the time (and distance) available for wind to deflect the plume.
2. Intercepts the plume at a point where centerline velocity is higher (more resistance to deflection).
3. Reduces the plume cross-section that must be captured.
4. Reduces the plume mass flow rate (lower CFM required).

Every 6 inches of mounting height reduction improves the critical wind speed for centerline exit by approximately 0.4 to 0.9 mph, depending on source type.

4.5 The Charcoal Wind Vulnerability

The charcoal kettle represents the most wind-vulnerable source type in the program. At 30" mounting height, the charcoal plume:

- Enters the CVP (bifurcated) regime at only 2 mph
- Begins losing 25% of plume mass at 3.7 mph
- Experiences centerline exit at 5.1 mph
- Is intermittently disrupted at 8 mph
- Is fully disrupted at 10 mph

These thresholds are 15-40% lower than for gas grill sources. The reason is the low convective heat release rate ($Q_c = 1.8$ kW for a kettle grill versus 8.2 kW for a medium gas grill), which produces a slower plume with less momentum to resist wind forcing.

The charcoal paradox identified in RB-001 is compounded by wind: the charcoal grill produces more contaminants per unit time but a weaker plume to carry them, and that weaker plume is more easily disrupted by wind. The combination makes charcoal grilling in wind-exposed locations the most

challenging ventilation scenario in the outdoor cooking domain.

Recommendation for charcoal installations: Mount the hood at 18 inches maximum. Use side panels and rear wall as standard. Limit charcoal grilling to Sheltered and Moderate wind exposure classes. In Exposed conditions (>7 mph), the charcoal plume cannot be reliably captured by any practical hood configuration.

4.6 Practical Design Hierarchy for Wind Mitigation

Based on the quantitative analysis, the following hierarchy ranks wind mitigation strategies by effectiveness per unit of cost and complexity:

1. **Reduce mounting height** (most effective, no cost). Every 6 inches lower improves critical wind speed by 0.5-0.8 mph and reduces CFM requirement by 22-44%. Mount as low as safety codes, head clearance, and operational access permit.
2. **Install side panels** (highly effective, low cost). Side panels on both sides recover 15-20% capture efficiency at 5 mph, equivalent to reducing wind exposure by one class. Cost: \$100-400 for aftermarket panels.
3. **Position against a wall** (highly effective, zero incremental cost if layout permits). A rear wall reduces effective wind at the plume by 60-80%. If the patio layout allows placing the grill against a wall with the cook facing outward, this is the most effective single strategy.
4. **Orient hood for wind** (free). Align the hood's long axis parallel to prevailing wind. Improves critical wind speed by 30-55% for rectangular hoods.
5. **Increase overhang** (moderately effective, moderate cost). Adding 6-12 inches of hood overhang compensates for approximately 3-5 mph of wind. Beyond this, the required overhang becomes impractically large.
6. **Increase CFM** (partially effective, moderate to high cost). Increasing CFM by 50-100% above still-air requirements provides partial compensation at 5-8 mph wind but cannot fully recover capture. May require upgrading blower, increasing duct size, and accepting higher noise.
7. **Full enclosure** (most effective for severe wind, highest cost). Three- or four-sided wind enclosure eliminates wind at the cooking surface, enabling Sheltered-class performance in any external wind condition. Required for Exposed and Severe class installations.

4.7 Impact on Downstream Topics

This paper's findings directly affect the following downstream research:

RB-007 (Failure Modes): Wind is identified as the primary cause of outdoor hood capture failure. The failure mechanism is geometric displacement of the plume beyond the **Capture Envelope**, not aerodynamic insufficiency of the hood's suction. The critical wind speed tables (Section 3.4) define the wind-speed boundaries for each failure threshold.

RB-008 (CFM Requirements): RB-008 adopts class-representative wind correction factors (Sheltered 1.3, Moderate 1.6, Exposed 2.5, or 1.8 with side panels) for its consolidated CFM tables; the speed-resolved factors in Table 4.3 apply when the wind speed at cooking height is measured. The RB-003 preliminary factors are superseded.

RB-009 (Side Panel Effectiveness): The quantitative framework for side panel analysis (Section 3.9.1) provides the foundation for detailed panel geometry optimization. The CFM-equivalent improvement values and wind class reduction metrics provide the performance targets.

RB-011 (Grease Aerosol Transport): Wind-driven plume escape carries grease aerosol beyond the hood, depositing it on surrounding surfaces. The deflection tables quantify the spatial distribution of escaped plume material as a function of wind speed and direction.

5. Knowledge Gaps or Opportunities

5.1 Well-Established Knowledge

The following aspects of wind-plume interaction are well-established in the atmospheric dispersion and fluid dynamics literature and form a reliable basis for the analysis in this paper:

1. **The Briggs plume rise equations** are validated against decades of atmospheric dispersion observations from power plant stacks and are accepted as standard engineering tools. The adaptation to small-scale cooking plumes introduces uncertainty but the physical framework is sound.
2. **The counter-rotating vortex pair (CVP) structure** in jets and plumes in crossflow is extensively documented in experimental and computational studies. The qualitative behavior (onset at Fr approximately 0.5, full development at Fr approximately 1.0) is robust.
3. **Gust factor relationships** in the atmospheric surface layer are well-characterized. The values used ($G = 1.5$ - 2.0) are standard meteorological practice.
4. **The wind profile logarithmic law** for the surface layer is well-established for typical terrain categories. The reduction from 10-meter reference speed to cooking-height speed (factors of 0.35-0.64) is reliable for the terrain categories considered.

5. **The physical principle that plume edge velocity is negligible compared to even light wind** is a direct consequence of the Gaussian radial velocity profile established in RB-001 and RB-003 and is not subject to significant uncertainty.

5.2 Areas of Moderate Uncertainty

1. **The calibrated deflection formula.** The formula $\Delta x = 0.35 * U_w * z / u_o(z)$ is calibrated against the RB-003 preliminary estimates and is consistent with the Briggs framework, but it has not been validated against measurements of outdoor cooking plumes specifically. The calibration constant $C_d = 0.35$ could range from 0.25 to 0.50 depending on source geometry, ambient turbulence, and the degree to which the hood's suction field deflects the plume. This introduces approximately 30-40% uncertainty in the deflection values.
2. **The wind-enhanced entrainment model.** The formula $\alpha_{eff} = \alpha_o * (1 + 0.7 * Fr)$ is a simplified representation of a complex three-dimensional process. The coefficient 0.7 is an order-of-magnitude estimate. Published values for the crosswind entrainment enhancement range from 0.3 to 1.5 depending on the flow configuration. The plume widening estimates in Section 3.5 carry corresponding uncertainty.
3. **The disruption thresholds.** The Froude number boundaries between coherent, intermittent, and disrupted regimes ($Fr = 1.5$ and 3.0) are approximate. Experimental observations of jets in crossflow show that the transition depends on Reynolds number, source geometry, and turbulence characteristics in addition to Froude number. The transitions are gradual, not sharp.
4. **Side panel and rear wall effectiveness estimates.** The wind reduction factors for side panels (40-60% effective wind reduction) and rear walls (60-80%) are based on general wind engineering principles for building sheltering. They have not been measured for the specific geometry of outdoor cooking installations with hoods.
5. **The capture efficiency percentages** in Tables 3.10 and elsewhere are engineering estimates, not calculated from validated models. The actual capture efficiency in any specific wind condition depends on the detailed three-dimensional flow field around the hood, grill, cook, and surrounding structures — which can only be determined by CFD simulation or experimental measurement.

5.3 Knowledge Gaps Requiring Further Research

1. **No experimental validation of outdoor cooking plume deflection in crosswind.** The entire analysis rests on adaptation of fire plume and atmospheric dispersion correlations. No published study has measured the deflection of a barbecue grill plume in controlled crosswind

conditions. A wind tunnel or outdoor field experiment measuring plume trajectory (via schlieren imaging, particle image velocimetry, or tracer gas) at representative wind speeds would provide critical validation.

2. **CFD modeling of hood-plume-wind interaction.** The coupled system of hood suction, buoyant plume, crosswind, and side panels involves complex three-dimensional flow patterns that are beyond the reach of the integral models used in this paper. Computational fluid dynamics (CFD) simulation — validated against the integral model predictions — would refine the deflection estimates, capture efficiency predictions, and side panel effectiveness values. Parametric CFD studies could also optimize panel height, panel angle, and hood lip geometry for wind resistance.
3. **Side panel optimization.** Section 3.9.1 treats side panels as simple flat barriers. In practice, panel height, angle, porosity, and shape can be optimized to maximize wind shielding while minimizing interference with user access and airflow to the combustion zone. No published study has optimized side panel geometry specifically for outdoor cooking hood performance.
4. **Wind direction variability.** This analysis assumes a steady, unidirectional wind. Real outdoor wind changes direction on timescales of minutes to hours. The plume responds to direction changes on timescales of seconds. The capture efficiency averaged over a variable-direction wind event is different from (and lower than) the efficiency computed for a steady wind from the most favorable direction. Statistical modeling of wind direction variability and its effect on time-averaged capture would improve the practical applicability of the wind exposure classification.
5. **Interaction between wind and food-generated aerosols.** Wind affects not only the thermal plume but also the transport of grease aerosol, smoke particles, and volatile organic compounds generated during cooking. The transport of these contaminants in a crosswind may differ from the thermal plume trajectory because the aerosol particles have inertia and settling behavior that differs from the gas phase. This coupling is relevant to RB-011 (Grease Aerosol Transport) and has not been analyzed.
6. **Dynamic wind events.** The gust analysis in Section 3.7 addresses the statistical properties of wind fluctuations but does not model the transient response of the plume to individual gust events. The plume has a finite response time (approximately $z / u_o = 0.3\text{--}1.0$ seconds), during which it transitions from its pre-gust trajectory to the gust-deflected trajectory. This transient behavior determines the fraction of time the plume spends beyond the hood boundary during gusty conditions and affects the time-averaged capture efficiency.
7. **Validation of the wind exposure classification.** The four-tier classification (Sheltered, Moderate, Exposed, Severe) is proposed based on the physical analysis in this paper. It has not been validated against field observations of actual outdoor cooking installations. A field survey

correlating reported ventilation effectiveness with measured wind conditions would validate or refine the classification boundaries.

6. Diagram Mapping Notes (Text Only)

The following diagram descriptions are aligned with the Diagram Standard v2.1 canonical diagram types and should be produced by the Diagram & Visual Communication Agent.

Diagram 6.1: Bent Plume Trajectory in Crosswind (Diagram Type 1 — Plume Profile)

Purpose: Illustrate the curved trajectory of the **Buoyant Cooking Plume** in crosswind, showing the deflection at various heights.

Content:

- Side-view cross-section showing cooking surface at bottom center, hood above
- Still-air plume shown as a vertical dashed outline (reference)
- Crosswind plume shown as a solid curved column bending to the right (downwind)
- Wind arrows (horizontal, left to right) at three heights showing uniform crossflow
- Plume centerline trajectory marked as a heavy dashed curve following $\Delta x(z)$
- Deflection distances marked at 18", 24", 30", 36", 48" heights
- Hood outline shown at 30" height, with the deflected plume partially exiting the hood on the downwind side
- The **Missed Plume Region** shaded in red: the portion of the plume beyond the hood edge on the downwind side
- The **Capture Envelope** shown as the hood's coverage area in green
- Labels: " $\Delta x = 12$ inches at 30" in 5 mph wind (Gas Medium)"
- Froude number annotation: " $Fr = U_w / u_o = 1.12$ at 30" — wind-dominated regime"
- Figure caption: "Figure 6.1: Bent plume trajectory for a Gas Grill Medium ($Q_c = 8.2$ kW) in a 5 mph crosswind. The plume centerline deflects 12 inches by the time it reaches the 30-inch hood. The downwind edge of the plume extends beyond the standard hood boundary, creating a Missed Plume Region where contaminant-laden gas escapes capture."

Diagram 6.2: Deflection Versus Wind Speed for Multiple Sources (Diagram Type 2 — Quantitative Chart)

Purpose: Provide a single reference chart showing plume deflection at 30-inch mounting height as a function of wind speed for all source types.

Content:

- X-axis: wind speed at cooking height (0 to 15 mph)
- Y-axis: plume centerline deflection at 30" height (0 to 50 inches)
- Eight curves, one per source type, color-coded by category (gas = blue, charcoal = red, wood = brown, pellet = green)
- Horizontal reference lines at: typical overhang (OH = 17" for gas medium), maximum hood overhang (OH = 20" for charcoal)
- Vertical reference lines at wind exposure class boundaries: 3 mph, 7 mph, 12 mph
- Annotation where curves cross the overhang lines: "Centerline exits hood at this wind speed"
- Shaded zones corresponding to wind exposure classes: Sheltered (green, 0-3), Moderate (yellow, 3-7), Exposed (orange, 7-12), Severe (red, >12)
- Figure caption: "Figure 6.2: Plume centerline deflection at 30-inch mounting height as a function of wind speed for all source types. Weak plumes (charcoal, pellet smoker low) deflect approximately twice as far as strong plumes (gas high-output) at any wind speed. The horizontal dashed line shows the typical hood overhang; curves crossing this line indicate centerline capture failure."

Diagram 6.3: Critical Wind Speed Map (Diagram Type 2 — Quantitative Chart or Heatmap)

Purpose: Provide a quick-reference visualization of the critical wind speed for centerline exit at each source type and mounting height.

Content:

- Heatmap with source types on the Y-axis and mounting heights on the X-axis
- Cell values: critical wind speed for centerline exit (mph) from Table 3.4
- Color scale: green (>10 mph — robust), yellow (7-10 mph — moderate), orange (5-7 mph — vulnerable), red (<5 mph — critical)
- Annotation: "Values are wind speed at cooking surface height for plume centerline to exit the RB-002 recommended hood"

- Figure caption: "Figure 6.3: Critical wind speed for centerline exit (centerline of Buoyant Cooking Plume exits the downwind edge of the recommended hood). Green cells indicate robust wind resistance (>10 mph); red cells indicate high vulnerability (<5 mph). Weak plumes at high mounting heights are most vulnerable."

Diagram 6.4: Froude Number Regime Map (Diagram Type 2 — Heatmap)

Purpose: Show the Froude number (and corresponding plume regime) at each combination of source type, mounting height, and wind speed.

Content:

- For a fixed mounting height (30"), show a matrix of source type vs. wind speed
- Cell colors: blue ($Fr < 0.3$, buoyancy-dominated), green ($0.3-1.0$, transitional), yellow ($1.0-1.5$, wind-dominated), orange ($1.5-3.0$, intermittent/disrupted), red (>3.0 , fully disrupted)
- Fr value displayed in each cell
- Annotations marking regime boundaries
- Figure caption: "Figure 6.4: Froude number regime map at 30-inch mounting height. The Froude number $Fr = U_{wind} / u_{plume}$ determines whether buoyancy (blue) or wind (red) controls the plume trajectory. At $Fr > 1.5$, the plume loses coherent structure and capture becomes unreliable. Weak sources (charcoal, pellet smoker low) enter the disrupted regime at lower wind speeds than strong sources."

Diagram 6.5: Mitigation Strategy Comparison (Diagram Type 4 — Comparative)

Purpose: Visually compare the effectiveness of different wind mitigation strategies for a Gas Grill Medium at 30 inches in 5 mph wind.

Content:

- Five side-by-side configurations, each showing the same source and hood:
 1. Standard hood, no mitigation: plume deflected, ~70% capture
 2. Standard hood + side panels: plume partially sheltered, ~90% capture
 3. Standard hood + side panels + rear wall: plume sheltered, ~95% capture
 4. Wider hood (69"): plume deflected but within hood, ~85% capture
 5. Standard hood + 80% more CFM: plume still deflected but edge capture improved, ~82% capture

- Each configuration annotated with estimated capture efficiency, cost estimate, and complexity rating
- Color-coded efficiency bar below each: green (>90%), yellow (80-90%), orange (70-80%)
- Figure caption: "Figure 6.5: Comparison of wind mitigation strategies for a Gas Grill Medium at 30-inch mounting height in 5 mph wind. Side panels plus rear wall provide the best capture recovery (95%) at moderate cost. Increasing CFM alone provides limited improvement because the problem is geometric (plume displacement), not aerodynamic (insufficient suction)."

Diagram 6.6: Wind Exposure Classification Decision Guide (Diagram Type 5 — Decision/Flow)

Purpose: Provide a practical decision guide for classifying a site's wind exposure and selecting appropriate mitigation.

Content:

- Decision tree starting with "What is the typical wind speed at cooking surface height?"
- Branch 1: < 3 mph --> Sheltered class --> Standard hood per RB-002/RB-003
- Branch 2: 3-7 mph --> Moderate class --> Add side panels; use $K_{CFM} = 3.68+$; mount at 30" max
- Branch 3: 7-12 mph --> Exposed class --> Full side panels + rear wall; $K_{CFM} = 6.0+$; mount at 24" max for gas, 18" for charcoal/pellet
- Branch 4: > 12 mph --> Severe class --> Wind enclosure required; postpone cooking if no enclosure
- Secondary decision: "Is prevailing wind direction consistent?" --> Yes: orient hood long axis parallel to wind; No: use symmetric overhang margins
- Figure caption: "Figure 6.6: Wind exposure classification decision guide. Start by estimating wind speed at cooking height (approximately 55-65% of weather-reported wind for suburban settings). Follow the decision tree to determine wind exposure class and required mitigation measures."

Diagram 6.7: Plume Cross-Section Evolution with Increasing Froude Number (Diagram Type 1 — Schematic)

Purpose: Show how the plume cross-section changes from circular Gaussian to CVP kidney shape to disrupted cloud as wind speed increases.

Content:

- Four cross-section views (looking down from above) of the plume at 30" height for Gas Medium:
 1. $Fr = 0$ (still air): circular Gaussian, centered under hood

2. $Fr = 0.5$ (2 mph): slightly elliptical, minor offset, CVP beginning
 3. $Fr = 1.0$ (5 mph): kidney-shaped CVP, offset 12" from center, wider than still-air diameter
 4. $Fr = 2.0$ (10 mph): disrupted, bifurcated lobes, mostly outside hood boundary
- Hood outline (57" x 53" rectangle) shown for reference in each panel
 - Contaminant concentration indicated by shading density
 - Wind direction arrow in each panel
 - Figure caption: "Figure 6.7: Evolution of the Buoyant Cooking Plume cross-section at 30-inch hood height as the crosswind Froude number increases from 0 (still air) to 2.0 (10 mph wind). The plume transitions from a circular Gaussian column to a kidney-shaped counter-rotating vortex pair to a disrupted, bifurcated cloud. The hood outline (rectangle) shows how the plume progressively escapes the Capture Envelope as wind increases."

Appendix A: Calculation Parameters

A.1 Source Parameters from RB-001

Source Type	Q_c (kW)	D_{eff} (m)	z_o (m)
Gas Grill — Small	5.1	0.43	-0.30
Gas Grill — Medium	8.2	0.51	-0.37
Gas Grill — Large	12.3	0.58	-0.41
Gas Grill — High-Output	16.4	0.65	-0.44
Charcoal Kettle	1.8	0.56	-0.47
Wood-Fired	7.6	0.50	-0.36
Pellet Smoker Low	1.5	0.45	-0.38
Pellet Smoker High	5.7	0.45	-0.30

A.2 Still-Air Plume Velocities from RB-001 Table 3.5

Height	Gas Small	Gas Med	Gas Large	Gas High	Charcoal	Wood	Pellet Low	Pellet High
18"	2.01	2.30	2.60	2.83	1.45	2.25	1.38	2.06
24"	1.85	2.12	2.39	2.60	1.33	2.07	1.27	1.90
30"	1.74	1.99	2.25	2.44	1.25	1.94	1.19	1.78
36"	1.64	1.88	2.12	2.31	1.18	1.84	1.13	1.68
48"	1.49	1.71	1.93	2.10	1.07	1.67	1.02	1.53

A.3 Hood Overhang from RB-002 Tables 3.6a-k ($K \approx 1.38$)

Source Type	OH at 18"	OH at 24"	OH at 30"	OH at 36"	OH at 48"
Gas Small	13"	14"	16"	18"	22"
Gas Medium	13"	15"	17"	19"	23"
Gas Large	12"	14"	16"	18"	22"
Gas High-Output	11"	13"	15"	17"	21"
Charcoal Kettle	16"	18"	20"	22"	26"
Wood-Fired	13"	15"	17"	18"	22"
Pellet Smoker Low	12"	14"	16"	18"	22"
Pellet Smoker High	12"	14"	16"	18"	22"

A.4 Standard Ambient Conditions

Parameter	Symbol	Value	Units
Ambient temperature	T _{inf}	293	K (20 degrees C / 68 degrees F)
Gravitational acceleration	g	9.81	m/s ²
Ambient air density	rho _{inf}	1.20	kg/m ³
Specific heat of air	c _p	1.00	kJ/(kg*K)
Entrainment coefficient (still air)	alpha _o	0.11	dimensionless
Deflection calibration constant	C _d	0.35	dimensionless
Gust factor (standard)	G	1.7	dimensionless
Gust factor (exposed)	G _{exposed}	2.0	dimensionless

A.5 Wind Speed Conversion Reference

Description	mph	m/s	ft/min	km/h
Barely perceptible	1	0.45	88	1.6
Light breeze	2	0.89	176	3.2
Gentle breeze	5	2.24	440	8.0
Moderate breeze	8	3.58	704	12.9
Fresh breeze	10	4.47	880	16.1
Strong breeze	15	6.71	1320	24.1

Appendix B: Glossary Terms Used in This Paper

All terms below are used as defined in Glossary v1.1 of the Outdoor Ventilation Standard.

- Buoyant Cooking Plume** — The thermally driven column of heated gas, combustion byproducts, and entrained ambient air rising from an outdoor cooking source. Characterized in RB-001.
- Wind-Affected Plume Behavior** — The modification of plume trajectory, cross-section, entrainment rate, and structural coherence caused by ambient wind. The central subject of this paper.

3. **Capture Envelope** — The region of space that the hood must cover to intercept the plume and achieve reliable capture. In crosswind, the Capture Envelope is displaced downwind and widened relative to the still-air position.
 4. **Missed Plume Region** — The portion of the plume cross-section that passes beyond the hood's physical boundary and escapes capture. Wind increases the Missed Plume Region by deflecting the plume and widening it asymmetrically.
 5. **Open-Boundary Dilution** — The irreversible dispersal of plume gas into the ambient atmosphere once it escapes the hood. In an outdoor environment, there is no recirculation, ceiling jet, or room pressurization to recover escaped plume gas.
 6. **Plume Interception Plane** — The horizontal plane at hood height where the hood intercepts the rising plume. In crosswind, the plume centerline at the Plume Interception Plane is offset from the source centerline by the deflection distance δ_x .
 7. **Effective Capture Area** — The portion of the hood's face area that actively contributes to plume ingestion. In crosswind, the Effective Capture Area on the windward side may extend beyond the plume boundary (wasting suction on ambient air), while the Effective Capture Area on the downwind side may be insufficient to reach the deflected plume.
 8. **Velocity Decay** — The reduction of plume centerline velocity with height, following the $z^{-1/3}$ power law. Wind-enhanced entrainment accelerates velocity decay beyond the still-air rate.
 9. **Entrainment Zone** — The plume boundary region where ambient air is drawn into the plume by turbulent engulfment. Crosswind increases the entrainment rate and widens the Entrainment Zone, particularly on the windward side.
 10. **Near-Field Plume Region** — The combustion zone immediately above the fuel bed where the plume is forming and the Boussinesq approximation is invalid. Wind effects in the Near-Field Plume Region can disrupt flame attachment and alter the initial plume structure, but these effects are beyond the scope of this paper.
 11. **Momentum-Limited Capture** — The condition where the hood's exhaust-induced suction is insufficient to ingest the full plume mass flow or resist wind-driven escape of the plume periphery. In crosswind, Momentum-Limited Capture manifests primarily as edge capture failure on the downwind side.
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References

1. Briggs, G.A. (1969). *Plume Rise*. USAEC Critical Review Series, TID-25075. U.S. Atomic Energy Commission.
2. Briggs, G.A. (1975). "Plume Rise Predictions." Chapter 3, *Lectures on Air Pollution and Environmental Impact Analysis*, American Meteorological Society.
3. Briggs, G.A. (1984). "Plume Rise and Buoyancy Effects." Chapter 8, *Atmospheric Science and Power Production*, ed. D. Randerson, DOE/TIC-27601.
4. Fric, T.F. and Roshko, A. (1994). "Vortical structure in the wake of a transverse jet." *Journal of Fluid Mechanics*, 279, pp. 1-47.
5. Mahesh, K. (2013). "The interaction of jets with crossflow." *Annual Review of Fluid Mechanics*, 45, pp. 379-407.
6. Karagozian, A.R. (2014). "The jet in crossflow." *Physics of Fluids*, 26, 101303.
7. Hewett, T.A., Fay, J.A., and Hoult, D.P. (1971). "Laboratory experiments of smokestack plumes in a stable atmosphere." *Atmospheric Environment*, 5, pp. 767-789.
8. Fackrell, J.E. and Robins, A.G. (1982). "Concentration fluctuations and fluxes in plumes from point sources in a turbulent boundary layer." *Journal of Fluid Mechanics*, 117, pp. 1-26.
9. Davidson, G.A. (1986). "Gaussian versus top-hat profile assumptions in integral plume models." *Atmospheric Environment*, 20, pp. 471-478.
10. Contini, D. and Robins, A.G. (2001). "Water tank measurements of buoyant plume rise and structure in neutral crossflows." *Atmospheric Environment*, 35, pp. 6105-6115.
11. Anfossi, D., Ferrero, E., Brusasca, G., Marzorati, A., and Tinarelli, G. (1993). "A simple way of computing buoyant plume rise in a Lagrangian stochastic dispersion model." *Atmospheric Environment*, 27A, pp. 1443-1451.
12. Webster, H.N. and Thomson, D.J. (2002). "Validation of a Lagrangian model plume rise scheme using the Kincaid data set." *Atmospheric Environment*, 36, pp. 5031-5042.
13. Morton, B.R., Taylor, G.I., and Turner, J.S. (1956). "Turbulent gravitational convection from maintained and instantaneous sources." *Proceedings of the Royal Society A*, 234, pp. 1-23.
14. Heskestad, G. (2016). "Fire Plumes, Flame Height, and Air Entrainment." Chapter 13, *SFPE Handbook of Fire Protection Engineering*, 5th ed. Springer.

15. Gifford, F.A. (1961). "Use of routine meteorological observations for estimating atmospheric dispersion." *Nuclear Safety*, 2, pp. 47-51.
16. Pasquill, F. and Smith, F.B. (1983). *Atmospheric Diffusion*, 3rd ed. Ellis Horwood.
17. Fisher, H.B., List, E.J., Koh, R.C.Y., Imberger, J., and Brooks, N.H. (1979). *Mixing in Inland and Coastal Waters*. Academic Press.
18. Hunt, J.C.R. (1994). "Atmospheric jets and plumes." In *Recent Research Advances in the Fluid Mechanics of Turbulent Jets and Plumes*, eds. P.A. Davies and M.J. Valente Neves, Kluwer Academic.

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